

Water Wars

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Abstract

How does the geography of water resources shape violent conflict? We assemble a georeferenced panel spanning the entire African continent from 1997 to 2023, combining detailed data on hydrology and conflict. Leveraging the spatial inequality in surface water resources and the structure of the river network, we show that heat shocks trigger conflict in nearby water-rich areas and that they propagate upstream along rivers, but not downstream. These effects are more pronounced when shocks cross ethnic boundaries along the river, in countries with weak national institutions, and in areas with persistent heat and declining water availability. Our findings highlight the geography of water resources as a key mechanism through which climatic shocks propagate into violence.

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1 — Introduction

Water resources are essential for human life and economic activity. Estimates suggest that half of the world's population experiences at least one month per year without sufficient water (Mekonnen and Hoekstra 2016), and climate change is likely to exacerbate this situation. Crucially, water is distributed unevenly in space and flows across political and ethnic boundaries. This geography shapes where communities can access water when climatic shocks raise demand. Such shocks may strain the institutions and informal arrangements that govern water access (Ostrom 1990). When cooperation fails, disputes over access may escalate into violence. These concerns have led scholars and policymakers to warn about the potential for *water wars* (Gleick 1993; United Nations 2023).¹ Yet systematic evidence on whether and how the geography of water resources shapes violent conflict remains limited.

In this paper we fill this gap by investigating empirically the role of surface water resources in shaping the occurrence of violent conflict in Africa from 1997 to 2023. Leveraging high-resolution data on hydrography, weather, and conflict, we show that the geography of water is a key mechanism through which climatic shocks diffuse across space and translate into violence. First, heat shocks raise conflict in nearby water-rich areas. Second, the river network structure generates an asymmetry in their diffusion: conflict increases with heat downstream, but not upstream. Water, under unfavorable climatic conditions, can induce a *conflict resource curse*.

Two facts about water motivate our research design and its two main empirical tests. First, water resources are distributed unequally in space. A temperature shock which raises the demand for water can push affected groups to seek access to it in nearby, water-rich locations. This logic motivates our first testable hypothesis: conflict should rise in a water-rich location when its lateral neighbors, defined as nearby areas not connected to it along a river, experience a temperature shock. Second, water flows along rivers, which creates an asymmetry. A downstream group that needs more water during a hot year must fight upstream to get it, while an upstream group can instead extract more of the flow locally. This logic motivates our second test: conflict along a river should respond more strongly to temperature shocks in downstream locations than in upstream ones.

The African continent is an ideal context to test these hypotheses. The economy is largely dependent on farming and pastoralism, and water from rivers and lakes can be used for these economic activities. For instance, farmers construct irrigation channels from rivers or practice recession agriculture on lands enriched by river sediments. Similarly, pastoralists exploit these water bodies as drinking points for their livestock. Property rights over land and water are weak,

1. For instance, in 2001, then UN Secretary General Kofi Annan stated that “*fiere competition for fresh water may well become a source of conflict and wars in the future*”.

and formal mechanisms to resolve competing claims are largely absent.

We bring these hypotheses to the data using cells of $0.5^\circ \times 0.5^\circ$ in latitude and longitude as units of observation. We measure climatic shocks using high-resolution temperature data from ERA5 and the incidence of conflict using geocoded event data across all African countries from Armed Conflict Location & Event Data project (ACLED), which provides details on the date, location, and type of conflicts. We rely on detailed hydrological data to measure the distribution of water resources and to determine, for each pair of cells, whether they are connected along the river network (*upstream* or *downstream*) or not (*lateral*).

Our empirical strategy leverages idiosyncratic variation in each cell's exposure to temperature in its neighborhood, defined as all surrounding cells within a 200 km radius. The first test asks whether the violence caused by high temperature in lateral neighbors is amplified in water-rich cells. Our preferred measure of water richness is *water discharge*, representing the average water flow through a cell.² The second test restricts attention to cells along a river and compares shocks originating upstream to shocks originating downstream. By employing geographically disaggregated data, in both tests we can include grid-cell fixed effects, to account for local time-invariant factors, and country-year fixed effects, to control for common macro-level factors that vary by country and year. Our approach also allows us to control for any direct effects of temperature occurring in the grid-cell itself.

The first main result is that exposure to temperature shocks increases the likelihood of conflict differentially more in locations that are water rich. Our preferred specification implies that when a cell is exposed to a one degree increase in temperature in the lateral neighborhood, the likelihood of conflict is 3.3 percentage points higher for a cell at the 75th percentile of water discharge compared to one at the 25th percentile, corresponding to 36% of the conflict incidence mean.

The second main result is that the structure of the river network plays a crucial role in transmitting the effects of temperature shocks on conflict. We find that the effect of temperature shocks on conflict is stronger when the shock occurs in a downstream cell, compared to an upstream cell: a one degree increase in downstream temperature raises the likelihood of conflict by 2.1 percentage points (21% of the mean), while the corresponding upstream effect is an order of magnitude smaller and not statistically significant.

As water richness and river presence can be correlated with other factors that may influence conflict, we show that our results are robust to controlling for a wide range of cell characteristics, including population density, urbanization, land suitability for agriculture, presence of mines, distance to ethnic or national borders, and distance from the coast. Moreover, we show that the

2. To validate that our measure of water richness is indeed capturing availability of resources that local communities can use for agriculture, we show that high temperature reduces agricultural productivity, as measured by gross dry matter productivity, but that this effect is attenuated in cells with higher water discharge.

findings are robust to spatial correlation in the error term, alternative measures of water richness or temperature shocks, and alternative neighborhood definitions.

Given that water is mostly used for agriculture in this context, we dig deeper into the type of conflict and classify violence with agricultural roots using the short descriptions of each conflict event. We further distinguish between two types of agricultural conflict: those related to competition over land and water (*factor* conflict) or over crops and livestock (*output* conflict). We find that the effects are larger when focusing on agricultural conflict, and that both types contribute, consistent with climatic shocks pushing groups to contest both the water itself and the output it sustains (McGuirk and Burke 2020).

We also investigate the role of water resources in shaping farmer-herder conflict (McGuirk and Nunn 2025; Eberle et al. 2025). By distinguishing between shocks originating in agricultural and in pastoralist territories, we find that both shocks trigger conflict in nearby water-rich cells, but the up-downstream asymmetry is driven by shocks in agricultural neighbors. This suggests that our results capture not only pastoralists migrating onto farmers' land (McGuirk and Nunn 2025), but also a distinct channel in which farmers themselves invade other communities' territory or fight to appropriate agricultural output and to prevent upstream over-extraction.

In the second part of the paper we turn to investigate which factors exacerbate conflict risk. We start by studying whether differences in ethnicity or nationality increase the probability that cooperation over shared water breaks down. Along the river network, we find that the effect of climatic shocks is stronger when they originate in a different ethnic group's territory.³ By contrast, we find no comparable difference when shocks originate in a different country. These findings are consistent with the high cooperation costs due to ethnic grievances documented in several contexts (Alesina and La Ferrara 2005; Hjort 2014) and with the predominant role of traditional leaders in land governance (Goldstein and Udry 2008; Boone 2014).

We then ask whether formal institutions attenuate conflict risk. Governments can help communities cope with adverse shocks through safety nets and infrastructure, mediate disputes between competing groups, and enforce a monopoly of violence that prevents disputes from escalating. Yet some governments lack the capacity to do so, and others lack the will, since acting would divert resources from their political base in urban centers. We therefore explore heterogeneity across three dimensions of institutional quality: the ability of the government to exert control over the entire territory, its effectiveness at delivering public services, and the degree of political inclusion of rural communities. We find that heat shocks' effects on violence are attenuated in countries which score higher along these dimensions.

Lastly, we move beyond transitory shocks and ask whether the long-run changes associated

3. This difference is statistically significant for the up-downstream channel; for the lateral channel we do not detect a differential effect depending on whether the shock originates in the same or a different ethnic group.

with climate change exacerbate the risk of violence. These changes are likely to take two forms: persistent warming, with higher average temperatures, and a lasting decline in surface water availability. Both forces may make cooperation harder to sustain. But they may also incentivize communities to invest in institutions that enforce it (Haseeb 2024) or in adaptation strategies that reduce their exposure to heat.

We explore the consequences of both forms of long-run change. To capture persistent warming, we replicate our main analysis using the rolling average of temperature over different time-windows, and find that the effects are stronger under prolonged exposure to high temperatures. To capture long-run changes in water availability, we leverage the full length of our time-series data, and find that the effect of hot years on conflict is stronger in areas where water has diminished over the past forty years. So far, then, we see no evidence of improved informal cooperation or of adaptation, which suggests that the conflict we document may become a more urgent issue with climate change.

Related Literature This paper contributes to the literature on the relationship between natural resources and violence (Dube and Vargas 2013; Berman et al. 2017; Adhvaryu et al. 2021; Vanden Eynde and Vargas 2025). Several authors across the social sciences have discussed the possibility that water resources can lead to violence (Gleick 1993; Homer-Dixon 1994), but there is limited empirical evidence on the occurrence of *water wars*. One exception is Gleditsch et al. (2006), who provide quantitative evidence that pairs of countries connected along the same river and with an upstream and downstream relationship are more likely to engage in interstate conflict.⁴ To the best of our knowledge, we are the first to provide evidence on how the spatial distribution of surface water resources also shapes intrastate violence. Moreover, by using granular data covering the entire African continent, we are able to conduct the analysis at a geographically disaggregated level to reduce concerns about potential confounders.

More broadly, our paper is closely related to the strand of this literature examining cooperation and disputes over access to groundwater and irrigation infrastructure (Sekhri 2014; Detges 2014; Döring 2020; Haseeb 2024; Couttenier et al. 2025). Two papers are particularly related to our work. First, Couttenier et al. (2025) show that, on the African continent, shallow groundwater exacerbates conflict risk during periods of high temperature. Our findings provide complementary evidence on the role of surface water, which is more widely used than groundwater for agriculture in this context and, by flowing along rivers, creates a natural interdependence among locations. This feature underscores the importance of considering spatial spillovers from temperature shocks. Second, Haseeb (2024) provides evidence from Pakistan on the relevance of water diversions

4. While the original paper finds results for country pairs connected by a transboundary river in general, later work by some of the same authors has shown that only the upstream and downstream connection seems to matter (Brochmann and Gleditsch 2012).

along irrigation canals as a way for farmers to cope with climatic shocks, and on how informal institutions capable of enforcing cooperation emerge in response to long term declines in water availability. We provide complementary evidence by studying a different context, the entire African continent, and by focusing on extreme failures of cooperation, violent conflict, in access to river water rather than irrigation water.⁵

Because water resources are a key input for agriculture, we complement research that has focused on violence over agricultural commodities and land control (McGuirk and Burke 2020; Berman et al. 2021; Le Rossignol et al. 2024). Relatedly, we speak to recent quantitative studies on conflict dynamics involving pastoralist groups (McGuirk and Nunn 2024, 2025; Eberle et al. 2025; Jensen et al. 2025; Gehring and Schaudt 2026). In particular, our findings suggest that water resources play an important role in shaping the links between climatic shocks and violence between farmers and herders studied in McGuirk and Nunn (2025) and Eberle et al. (2025).

Our research also contributes to the literature on climate and conflict by presenting new evidence that identifies a precise mechanism through which climate change (Hsiang and Burke 2014; Burke et al. 2015; Burke et al. 2024) and climatic shocks (Miguel et al. 2004; Couttenier and Soubeyran 2014; Sarsons 2015; Almer et al. 2017; Breckner and Sunde 2019; Unfried et al. 2022) influence local violence. We focus on the effects of temperature, which is increasing in Africa as a result of climate change, and show that one mechanism through which heat can affect conflict is by increasing competition over access to and control of surface water resources. Moreover, our results point to the interaction between short term shocks and long term declines in water availability as a factor that exacerbates conflict risk. This provides suggestive evidence on how climate change, by raising temperatures and altering the distribution of water resources, may amplify violence.

Our analysis also speaks to the spatial spillovers documented in existing research on climate and conflict (Harari and Ferrara 2018): we show that high temperature in one area leads to more conflict in nearby water-rich or upstream locations. By identifying this diffusion structure, we complement the specific mechanisms of conflict diffusion studied by König et al. (2017), Couttenier et al. (2024), McGuirk and Nunn (2025), and Guariso and Rogall (2026).

Outline The rest of the paper is organized as follows. Section 2 provides some background on how rivers and lakes' water is used for economic activity in rural Africa. In Section 3 we introduce our data sources and we detail how we build the variables used in the analysis. Section 4 describes the empirical strategy, and Section 5 presents the results of the paper. Finally, Section 6 concludes.

5. More specifically, our results on the asymmetry induced by the structure of river networks and on the role of ethnic differences in hampering cooperation parallel the findings in Haseeb (2024). However, our result on long term water decline exacerbating conflict risk suggests that, in our context, long term scarcity has not predominantly led to the emergence of institutions capable of enforcing cooperation.

2 – Context

Water is an essential resource for farming, pastoralism, and daily consumption. In rural Africa, where piped supply and large-scale irrigation are scarce, households depend heavily on rainfall and on surface water: rivers, streams, lakes, and seasonal floodplains. Because surface water is unequally distributed in space and flows along rivers, different communities face different costs in accessing it. We first illustrate how communities depend on surface water. We then show how formal and informal institutions allow different groups to cooperate in using it, and give examples in which cooperation instead fails and competition escalates into violence.

Economic value of surface water resources Securing land along a river grants access to water. For example, a common agricultural practice is flood-based farming, which capitalizes on the nutrient-rich soil deposited by river floods; a related practice uses inundation canals, where fields are irrigated through channels fed by seasonal high water in rivers (for more details see Puertas et al. 2021). Canals also carry river water to arid land. For instance, in Kenya paved canals drawing from the Turkwel River by the World Food Programme irrigate the fields of more than 45,000 farmers (World Food Programme 2023). Farmers located near rivers enjoy lower canal-construction costs, and canals are often paired with small dams that store water to regulate flow or hold it for the dry season. Surface water is just as central for pastoralists: rivers and lakes are natural watering points for livestock, and the land around them keeps its vegetation even in dry seasons.

Extraction of water for agriculture can reduce availability further downstream. A clear case is the Omo River, which flows from Ethiopia toward Lake Turkana in Kenya (Climate Diplomacy 2023c). Communities of the Lower Omo valley combine flood-recession agriculture with pastoralism, and both activities depend on the river's seasonal floods. Those floods became less frequent once irrigated sugar plantations were established upstream in Ethiopia, illustrating how upstream uses of surface water propagate downstream.

Cooperation and conflict over surface water resources The unequal distribution of water does not necessarily lead to conflict; groups often cooperate to share and manage it. Access to rivers and the water drawn from them is typically regulated by informal and formal institutions. Stambach and Kwayu (2021) describes the water-distribution system of the western Kilimanjaro region of Tanzania, where managers decide which day each household may divert the river flow onto its fields; in some communities the role is hereditary and customary, in others the manager is elected and answers to a regional water authority. This type of institution is not unique to Tanzania. Gomes (2006) describes the customary associations that run canal systems serving farmers and pastoralists in Somalia, monitoring compliance with water allocations, overseeing canal maintenance, and mediating disputes between communities. Formal institutions can play a

similar role: in Kenya's Naroosura area, a statutory Water Resource Users Association coordinates local canal committees that schedule irrigation and maintain the catchment (Agade et al. 2022). Similarly, social norms govern the relationship between farmers and pastoral groups: the former have usually granted pastoralists *corridors* to bring their animals to the river banks, and sometimes allowed them to use the fields after harvest. In the inland Niger delta of Mali, for instance, the customary Dina code long reserved livestock entry routes and set a fixed order of access to the river's flood pastures, keeping cultivated fields at a distance (Benjaminsen and Ba 2009); across the Sahel, some governments have more recently delimited transhumance tracks precisely to guarantee pastoralists' access to riverine water points (Cotula 2006).

While peaceful sharing is the norm, external pressures such as climate change and drought, land privatization, and large-scale irrigation can perturb these arrangements and at times escalate into violence. For instance, driven by climate change, pastoralists migrate earlier toward water-rich areas and clash with farmers whose crops are still standing (McGuirk and Nunn 2025). Heat and drought also push pastoralists to alter their routes, bringing them into competition with other pastoralists over the same water, in places where no sharing arrangement exists. In the borderlands of the Lower Omo and Turkana, communities searching for water and pasture have expanded their ranges, coming into closer proximity. Between 1989 and 2011, clashes between the Nyangatom, Daasanach and Turkana alone caused more than six hundred direct deaths (Climate Diplomacy 2023b). The same logic can also lead farmers to expand onto former grazing land: for instance in the Ségou region of Mali, the expansion of agriculture and the growth of irrigated schemes have pushed pastoralists off dry-season pastures and cut their access to the river (Goulden and Few 2011). In the same delta, paddy rice has expanded steadily at the expense of the flood pastures on which pastoralists depend, with about a quarter of them converted to rice fields since the 1950s (Benjaminsen and Ba 2009).

A second dynamic is the asymmetry between upstream and downstream users: those who control the upper reaches divert water in their favor and leave those below short. During heatwaves, upstream farmers extract more water for irrigation and reduce the flow reaching downstream groups. Downstream users may turn to violence, destroying irrigation works or threatening upstream farmers, especially where the state does not intervene. Agade et al. (2022) documents exactly this pattern along the Naroosura River in Kenya, where disputes between upstream and downstream users are frequent and often violent.⁶ A local police occurrence book recorded repeated incidents over a single year, involving both farmers and pastoralists, including demonstrations over lost downstream flow and the destruction of pumps and offices (Agade et al. 2022).

6. In the words of a pastoralist interviewed in that study: *"Farmers here use all the water ... that is when we come upstream to destroy dams here and burn generators ... we never have any problems with the people of Naroosura except during the drought period."*

Similar disputes are documented in Laikipia county, Kenya, and in Fayoum, Egypt (Nation 2023; Monitor 2022). Such tensions can escalate well beyond the local scale. Along the Senegal River, the Diama and Manantali dams raised the value of irrigable riverbank land in the 1980s; the ensuing scramble for that land, ignited by a clash between farmers and pastoralists over water and grazing, escalated into mass expulsions and the 1989 border war between Senegal and Mauritania (Parker 1991).

These examples point to two distinct ways in which competition over water resources can trigger violence. In the first, groups migrate from water-poor toward water-rich areas in search of water for their animals and land for cultivation, and clash with those already there. In the second, downstream groups confront upstream ones to secure a larger share of the flow. These two dynamics motivate the empirical strategy described in Section 4.

3 — Data

This section describes the data sources and the construction of the variables used in the analysis. Our empirical analysis is based on a geo-referenced, annual panel that divides the African continent into 10,202 grid cells. These grid cells have a size of $0.5^\circ \times 0.5^\circ$ degrees, equivalent to approximately $55 \text{ km} \times 55 \text{ km}$ at the equator. Throughout our analysis, the unit of observation is a cell-year pair. We first introduce our main data sources and then describe how we build the key variables used in the analysis. Full definitions and sources for all additional variables are provided in Appendix A, and Table A1 reports summary statistics.

3.1. Data Sources

Conflict Our study utilizes georeferenced conflict events from the Armed Conflict Location & Event Data Project (ACLED) covering the 1997–2023 period (Raleigh et al. 2010). The project aggregates event data from multiple sources, including regional and national media outlets, non-governmental organizations (NGOs), and humanitarian agencies. Each record includes the date and geographic coordinates of the event. We restrict our sample to events with high spatial precision.⁷ We focus exclusively on events categorized as “battles” and “violence against civilians,” thereby excluding less violent occurrences such as “riots” and “protests.” Using the date and spatial coordinates, we map each event to a specific grid cell-year pair and aggregate the data accordingly. We construct a binary conflict incidence indicator, which takes a value of 1 if at least one conflict event occurred within a given cell-year, and 0 otherwise. Additionally, leveraging the granular event descriptions provided by ACLED, we identify agriculture-related conflicts, distinguishing

7. Specifically, our analysis only includes events geolocated at the town level or higher (i.e., ACLED precision levels 1 and 2).

between disputes over factors of production (land and water) and those over agricultural output (crops and livestock). For robustness checks, we employ alternative georeferenced conflict data from the Uppsala Conflict Data Program (UCDP) (Sundberg and Melander 2013).

Hydrology In our analysis, we utilize river discharge reanalysis data from the Global Flood Awareness System.⁸ River discharge refers to the volume of water passing through a river cross-section per unit of time, measured in cubic meters per second. This dataset provides daily average river discharge on a global scale at a spatial resolution of $0.05^\circ \times 0.05^\circ$. Note that the water quantity reported in the dataset accounts for all types of surface water bodies, including lakes, ponds, rivers, and streams. For each grid cell, we extract the water discharge on the 15th day of each month and calculate the yearly average.

To incorporate information on the river network topology, we rely on the HydroBASINS dataset.⁹ This dataset provides shapefiles of drainage basins, which are globally consistent geospatial units frequently employed in environmental and hydrological studies.¹⁰ Each basin represents a topographic catchment area, such as a valley, that collects and channels precipitation. We assign each grid cell to a specific basin based on the amount of water present in the overlapping area. Next, leveraging the Pfafstetter coding system, we construct a matrix that maps the spatial relationships between each pair of cells along the river network.¹¹ This matrix enables us to identify whether any given pair of cells is connected upstream, connected downstream, or entirely unconnected.¹² By mapping these hydrological relationships to uniform square grid cells, we can employ the standard units of observation established in the conflict literature while simultaneously integrating the spatial structure of the river network. Finally, we incorporate data from Messenger et al. 2021 to identify the presence of perennial and seasonal rivers within our grid cells.

Temperature data Our main proxy for weather shocks is temperature, whose effect on conflict is among the most robust regularities in the climate-conflict literature: higher than usual temperatures depress agricultural output and raise the risk of localized violence across a wide range of settings (Burke et al. 2015; Eberle et al. 2025; Burke et al. 2024).

We draw temperature from the ERA5 database of the Copernicus Climate Change Service (Hersbach et al. 2023), a reanalysis that combines worldwide observations with a physical model to produce hourly temperature on a $0.25^\circ \times 0.25^\circ$ grid, which we aggregate to the cell-year level for

8. Accessible from Harrigan et al. 2020.

9. Part of the HydroSHEDS suite and accessible at <https://www.hydrosheds.org/>; for further details, see Döll et al. 2003.

10. The shapefiles are available at various levels of aggregation. We utilize level 7, as its average basin area most closely matches the size of our grid cells.

11. The Pfafstetter coding system is widely used in hydrology to determine upstream-downstream relationships between river basins; see, for example, Verdin and Verdin 1999.

12. See Appendix A.6 for further details on the construction of the river network matrix.

1997-2023. Reanalysis offers two advantages over weather data interpolated from ground stations. First, by assimilating continuous satellite telemetry into a physical model, ERA5 delivers objective, spatially complete, and uninterrupted measures. This is particularly relevant in developing regions, where the active weather-station network is sparse (Dell et al. 2014), and in contexts where armed conflict may impact the operability of ground stations (Harari and Ferrara 2018). Second, ERA5’s hourly resolution allows us to construct an alternative heat measure, the number of killing degree days in a year (Schlenker and Roberts 2009), which we use as a robustness check. Throughout, we demean temperature by the cell-specific sample mean so the variable captures deviations from local average conditions.

3.2. Variable Definitions

Neighborhood and exposure to temperature shocks For each cell of our grid, we define a neighborhood as all the cells within a 200 km radius.¹³ Figure A1 illustrates the construction on the Tana River in Kenya. The left panel superimposes the neighborhood on the river-basin shapefile, shaded by average discharge, with the reference cell in blue at the center. The right panel shows the resulting classification:¹⁴ green cells lie upstream of the reference cell, orange cells lie downstream, and gray cells fall within the neighborhood but have no upstream-downstream relationship with it. We refer to this last group as *lateral* cells.

We measure the exposure of a cell to temperature shocks as the average temperature in its neighborhood, excluding the cell itself. As we use demeaned temperature data, this variable captures how each cell is exposed to temperature anomalies in other cells within its neighborhood.

Water richness and on-river sample definition Our definitions of water richness are based on river discharge. Our preferred measure is Discharge (log), a continuous variable corresponding to the natural logarithm of the mean amount of freshwater present in a cell during the sample period 1997-2023 (see Figure 1).¹⁵ We also show results using time-varying measures of discharge. We define cells on a river as the ones with a positive number of kilometers of perennial or seasonal rivers in their main hydrographic basin and exhibits a strictly positive average water discharge over our sample period (see Figure A3).¹⁶

13. We explore the sensitivity of our results to alternative radii.

14. In Appendix Section A.6 we provide further details on how we construct this classification.

15. Because several cells have 0 discharge we build this measure by taking the logarithm of $1 + \text{Discharge}$

16. For each cell, we define its main basin as the one with the largest discharge.

4 – Empirical Strategy

Our objective is to investigate systematically how water resources shape local violence. Conflict can arise because groups compete over access to water or over the agricultural output that water resources sustain.

To visualize what we mean by water resources, in Figure 2 we show two cells in Kenya. The one in the left panel, just east of Lake Turkana, is essentially arid and has no water resources (it has a value close to 0 in our data on water discharge, sitting at the 3rd percentile in the sample distribution). The one in the right panel, depicting a portion of the Tana River, has several river and stream segments which make it rich in water resources (it has a value of about 2,490 m^3/s in our data on water discharge, sitting at the 88th percentile in the sample distribution). We expect to observe conflict related to water resources in the second cell, but not in the first one.

Our empirical strategy relies on a combination of cells' water richness and on the differential exposure to idiosyncratic climatic shocks. It consists of two separate empirical tests. First, we test whether conflict rises in a water-rich location when its neighbors experience a temperature shock. This is motivated by the fact that temperature shocks which raise the demand for water can push affected groups to seek access to it in nearby, water-rich locations. Second, we test whether conflict along a river increases more to temperature shocks in downstream locations than in upstream ones. This is motivated by the fact that, when hit by a temperature shock, upstream neighbors can extract more of the flow locally whereas downstream ones can only fight upstream to get it.

In what follows, we motivate in detail these two empirical tests. We first clarify why identifying the causal effect of water resources on violence requires exogenous shocks to the value of water. We then describe why temperature can generate such shocks, discuss the other channels through which temperature may affect violence, and show how these concerns motivate our focus on exposure to nearby temperature shocks rather than on local temperature alone. We then describe how spatial inequality in water availability and the structure of river networks yield two testable hypotheses. Finally, we present the estimating equations that we employ to test them and discuss potential threats to identification.

Identification problem Identifying the causal effect of water resources on violence requires addressing an obvious endogeneity concern: water-rich areas may differ systematically from water-poor areas along other dimensions that drive conflict, such as population density or land productivity. The ideal experimental design would be to randomly assign water resources across locations, an experiment that is evidently unavailable in this setting. Therefore, to address these endogeneity concerns, we follow the identification strategy developed in the recent economics of conflict literature, which exploits temporal variation in climatic or commodity price shocks

interacted with local cell characteristics (Dube and Vargas 2013; Berman et al. 2017; Berman et al. 2021; Eberle et al. 2025; Couttenier et al. 2025).

Water resources lack a market price. A natural approach is to leverage temperature shocks, which raise water demand and indirectly increase its shadow value through the food production it sustains.

Channels linking temperature and conflict Consistent with the empirical and theoretical literature, temperature shocks can affect the economic incentives to fight through three channels (Burke et al. 2015). First, heat lowers agricultural productivity, and hence the shadow wage of peaceful work: fighting becomes cheaper (*opportunity cost*). Second, by lowering agricultural productivity, heat makes food scarce: when food demand is inelastic, as is typical of subsistence agriculture, productivity shocks translate into higher shadow prices (Nath 2025; Hsiao et al. 2026). A higher shadow food price raises the value of water-supported output relative to the non-food costs of appropriation (*rapacity* over the factor of production). Third, when heat strikes the location where violence is observed, it reduces the quantity of agricultural output produced there while raising its shadow price. Local violence is therefore affected through both a quantity effect and a price effect; depending on which force dominates, heat can raise or lower the value of the lootable prize located there (*rapacity* over output).

These channels are not confined to the location of the shock. A growing literature documents that climatic and economic shocks propagate across space (Harari and Ferrara 2018; König et al. 2017; Couttenier et al. 2024). In our context the logic is immediate. The opportunity-cost channel is a *push* force: a group whose shadow wage has collapsed is willing to fight, and that fighting need not be local. The rapacity channels are a *pull* force. For local violence, a cell's own heat affects the local prize through both the quantity of output and its shadow price. For spillover shocks, by contrast, the struck neighbor's heat changes the origin-side valuation of food and water-supported output, while the target's own quantity is directly unchanged. Any empirical design that studies local shocks in isolation therefore misses part of the effect, and possibly most of it.

Local temperature and conflict Consider first the effect of a cell's own temperature shock on the conflict it experiences. The three channels collide. The shock pushes local groups to engage more in fighting, but part of that fighting can be directed elsewhere; at the same time it shrinks the local prize, so the cell attracts less violence from outside.¹⁷ The net effect is not signed. Nor does the ambiguity resolve in water-rich cells: water buffers local production, so the push is weaker, but it also protects the local prize, so the pull is stronger. Own temperature shocks therefore deliver no signed prediction, in general or in water-rich cells, and cannot discriminate between

17. Own heat raises the shadow price of the cell's own output but lowers its quantity, so the net effect on the value of the local prize is ambiguous. We assume the quantity effect dominates, so that own heat lowers the local prize.

the mechanisms we are after. We treat own temperature as a control throughout, so that it absorbs the direct local channels and our tests rest on the spillover variation.

Nearby temperature and conflict The picture changes when the shock hits a neighbor. A temperature shock in a nearby cell moves the origin-side forces: it lowers the neighbor's shadow wage and raises the neighbor's shadow price of food, while leaving the cell's own output directly unaffected. Because the neighbor now values food more highly, it also values more highly the water and output standing in the cell: the cell's attractiveness as a target rises even though its own quantities are unchanged. This increase in attractiveness is larger for cells endowed with more water resources, which hold a larger prize. For lateral neighbors, where peaceful diversion is unavailable, all these forces move in the same direction and imply that conflict in a water-rich cell should rise with the heat experienced by its neighbors. Our research design builds on this insight, and leverages spillovers in temperature shocks thanks to both variation in the origin of shocks and in the local characteristics of the cells exposed to them.¹⁸ Consistently with the relevance of these spillovers, in our same context Eberle et al. (2025) show that spikes in temperature in the ethnic homeland of a nomadic group tend to increase the likelihood that the group fights in other territories, especially in areas with water supply or suitable for agriculture.

As an illustration, in panel (a) of Figure 3 we display the same water-rich cell in Kenya of Figure 2, but we now also show its neighboring cells within 200 km. In panel (b) we show the temperature conditions in the same area during a normal year (2007), and in panel (c) and (d) we show the temperature conditions during hot years (2019 and 2016). One can easily see that in panel (c) the heat shocks are stronger in the north-western part of the neighborhood, whereas in panel (d) they are stronger in the south-eastern part. These examples illustrate how the same cell can be exposed to time-series variation in surrounding average temperature conditions, but also to time-series variation in the origin of these shocks.

The geography of water, temperature and conflict Two facts about water are key for our identification. First, water resources are distributed unequally. A temperature shock in one cell raises the incentives of affected groups to contest water access and agricultural output in nearby water-rich cells. This incentive is strongest for groups in water-poor cells: their production is least buffered against heat, and they have no water of their own to fall back on. Panel (a) of Figure 4 shows water discharge in the neighborhood of the water-rich cell, illustrating substantial spatial inequality in water resources. This variation allows us to test for differential effects of the same climatic shocks based on cell local characteristics, namely the water resource endowment.

18. Recent papers have leveraged spillovers in climate shocks to identify causal effects. McGuirk and Nunn (2025) leverage differential exposure to climatic shocks originating in either transhumant pastoralists or sedentary groups to document how climate change intensifies farmer-herder conflict. Gehring and Schaudt (2026) leverage differential exposure to climatic shocks across pastoralists with different insurance coverage to evaluate the effect of index-based insurance on violence.

Second, water flows one way along rivers, generating an asymmetry. When a neighboring group experiences a heat shock and seeks additional water, its position along the river determines whether it has a peaceful margin. If the shocked group is upstream of the target, it can intercept part of the flow before it reaches the target. If the shocked group is downstream, that water has already passed the target, so the group must move or fight upstream to secure it. To the extent that upstream groups meet a heat shock by diverting rather than fighting, the same shock should generate more conflict when it strikes downstream than when it strikes upstream. Panel (b) of Figure 4 shows how each cell in the neighborhood is related to the water-rich cell through the river network: some cells are upstream (green), some are downstream (orange), some are lateral (that is, not connected along the river; gray). This variation allows us to test whether the same cell responds differently to shocks depending on whether they originate upstream or downstream.

The first fact affects how a cell's exposure to climatic shocks in nearby areas not connected along the river network (*lateral* cells) translates into conflict, and how this depends on the spatial distribution of water resources. The second fact affects how a cell's exposure to climatic shocks in nearby locations along the same river (*upstream* and *downstream* cells) translates into conflict. Putting these two facts together, we can formulate two testable hypotheses.

- H1 A cell should see an increase in the likelihood of conflict if it is water rich and nearby lateral cells experience a temperature shock. The effect should be strongest when the target cell is water rich and the heated lateral neighbors are water poor.*
- H2 For a target cell located along a river, heat shocks to downstream neighbors should raise conflict at the target more than heat shocks to upstream neighbors.*

Both hypotheses align with the case studies discussed in Section 2. Next, we present the estimating equations which we use to test these hypotheses empirically.

Testing H1: Estimating equation We investigate the effect of exposure to nearby temperature shocks outside the river network on conflict in water-rich cells using a linear probability model:

$$y_{it} = \alpha_0 \text{Temperature}_{it}^{\text{Own}} + \alpha_1 \text{Temperature}_{it}^{\text{Lateral}} + \gamma \text{Temperature}_{it}^{\text{Lateral}} \times \text{Water Rich}_i + \mu_i + \mu_{ct} + \varepsilon_{it} \quad (1)$$

Where y_{it} is an indicator for conflict incidence in cell i during year t , Water Rich_i is a measure of water richness in a given cell, $\text{Temperature}_{it}^{\text{Own}}$ denotes the temperature in the cell, and $\text{Temperature}_{it}^{\text{Lateral}}$ captures temperature conditions in the neighborhood of a cell, excluding cells

along the same river.¹⁹ The coefficient of interest γ is therefore identified from how the conflict response to neighborhood heat varies with a cell's water richness. We include cell fixed effects μ_i and country-year fixed effects μ_{ct} to account for time invariant cell characteristics and country-wide shocks that might affect conflict. ε_{it} is the error term. In all specifications we estimate standard errors with a spatial HAC correction allowing for both cross-sectional spatial correlation and location-specific serial correlation, drawing on the method developed by Conley (1999) and recently applied in Eberle et al. (2025). We consider infinite serial correlation and use a radius of 200 km for the spatial kernel, corresponding to the main radius considered constructing the temperature exposure measure.

The parameter γ captures the differential effect of neighborhood temperature shocks on conflict in water-rich cells relative to non-water-rich cells. A positive estimate of γ indicates that, consistent with *H1* above, hotter years cause additional conflict in nearby water-rich cells. We also test whether the effects are stronger for shocks originating in *water poor* cells, that is in cells with less water than the reference cell.

Testing H2: Estimating equation We investigate the effect of exposure to nearby temperature shocks along the river network on conflict using a linear probability model:

$$y_{it} = \alpha_0 \text{Temperature}_{it}^{\text{Own}} + \gamma_1 \text{Temperature}_{it}^{\text{Up}} + \gamma_2 \text{Temperature}_{it}^{\text{Down}} + \mu_i + \mu_{ct} + \varepsilon_{it} \quad (2)$$

Where everything is like in Equation 1, except that $\text{Temperature}_{it}^{\text{Up}}$ and $\text{Temperature}_{it}^{\text{Down}}$ capture temperature conditions in neighboring cells along the same river, distinguishing between upstream and downstream neighbors. For this specification we restrict the sample to cells located along a river.

In line with *H2* above, our hypothesis is that temperature shocks originating from downstream cells are more likely to cause conflict than shocks originating from upstream cells; thus we expect $\gamma_2 > 0$ and $\gamma_2 > \gamma_1$.

Discussion The key identifying assumption is that, conditional on a rich set of fixed effects and on the main effects of temperature, the interaction between idiosyncratic fluctuations in neighborhood temperature and local water richness is uncorrelated with other determinants of conflict at the cell level.

It is important to note that our econometric specification accounts flexibly for many factors that have been previously studied in the conflict literature. The cell fixed effects capture all time-invariant determinants of conflict, such as geography, national boundaries, local institutions,

19. Because in our baseline specification Water Rich_i is time-invariant, its level is absorbed by the cell fixed effects, and it enters Equation 1 only through its interaction with lateral temperature

historical factors, and ethnic traits (Michalopoulos and Papaioannou 2016; Berman et al. 2017; Moscona et al. 2020; Arbatlı et al. 2020; Adhvaryu et al. 2021). The country-year fixed effects capture time-varying national-level factors such as changes in economic cycles, national political or legal institutions, country-level ethnic fractionalization and polarization, resource endowments, country-wide climatic trends, and international geo-political characteristics, all of which have been prominent in the cross-country literature on conflict (Miguel et al. 2004; Montalvo and Reynal-Querol 2005; Esteban et al. 2012). By directly controlling for temperature in the reference cell, we capture the direct effects of local temperature shocks on the opportunity costs, returns, or logistics of fighting, which have been documented in previous work (Burke et al. 2015; Harari and Ferrara 2018; Burke et al. 2024).

The main concern is that other characteristics of water-rich cells (e.g. soil fertility, population density) might lead to a differential response to neighborhood temperature shocks and thus confound the effect of interest. To address this concern, we show in Section 5.1 that results are robust to allowing neighborhood temperature shocks to have a differential effect in cells with other characteristics possibly related to both water richness and conflict.

5 — Results

Temperature, water resources, and agricultural productivity Before analyzing the impact of temperature shocks on conflict, we examine whether water resources indeed buffer the negative effects of temperature on agricultural productivity (as proxied by gross dry matter productivity). Table B1 shows that while higher temperature reduces agricultural productivity, greater water discharge attenuates the adverse effects of these shocks. A one-degree temperature shock lowers productivity by about 24% in a water-poor cell at the 25th percentile of discharge, but only about 5% in a water-rich cell at the 75th percentile.

Testing H1: Results Table 1 reports the results from Equation 1. Columns 1 and 2 estimate the effect of temperature shocks in laterally adjacent cells on conflict incidence. Lateral temperature shocks have no effect on conflict in column 1, while the interaction between lateral exposure and water richness in column 2 is positive and highly significant.²⁰

The result is consistent with temperature shocks triggering conflict in water-rich cells. To illustrate the magnitude, we compare the water-rich and water-poor example cells shown in Figure 2, whose discharge sits at the 88th and 3rd percentiles of the sample distribution. A one degree (one s.d.) increase in lateral temperature implies an increase in the probability of conflict

20. The direct effect of own-cell temperature is negative, though not statistically significant in this specification. As discussed in Section 4, our verbal framework does not sign this coefficient: own heat raises the local willingness to fight, but that fighting need not remain local, and it simultaneously shrinks the local prize, making the cell a less attractive target.

of 4.8 (2.2) percentage points at the water-rich cell's discharge relative to the water-poor cell's, corresponding to 53% (25%) of the dependent variable's mean. Using instead the 75th versus 25th percentile of the discharge distribution as a comparison, the same one degree (one s.d.) increase in lateral temperature implies an increase in the probability of conflict of 3.3 (1.5) percentage points, or 36% (17%) of the dependent variable's mean. Column 3 shows that the interaction is robust to using a time-varying, cell-year measure of discharge in place of the sample-period mean.

Column 4 unpacks the mechanism further, distinguishing between temperature shocks originating in neighboring cells that are relatively richer (Water-Rich) or poorer (Water-Poor) in water resources than the reference cell. The interaction with the reference cell's own discharge is positive and significant for shocks originating in water-poorer neighbors, but is essentially zero and insignificant for shocks originating in water-richer neighbors, showing that the effect of temperature shocks in water-poorer cells is driven by reference cells with greater water discharge.

Overall these results are consistent with H1: cells see an increase in the likelihood of conflict if they are water rich and the nearby lateral cells experience a temperature shock, with the effect being stronger if the lateral cells have less water resources than the reference cell.

Testing H2: Results Table 2 compares locations along rivers, which are arguably similar in their underlying characteristics, including water availability. Columns 1 and 2 show that exposure to downstream shocks leads to an increase in violence, whereas the same is not true for exposure to upstream shocks. Column 3 reports results for the full specification with both upstream and downstream temperature. Downstream temperature shock has a positive and significant effect while upstream shock's coefficient is smaller and not significant.²¹ In terms of magnitude, a one degree (one s.d.) increase in downstream temperature implies an increase in the probability of conflict of 2.1 (0.9) percentage points, or 21% (9%) of the dependent variable's mean; the corresponding upstream effect is an order of magnitude smaller, at 0.4 (0.1) percentage points, or 4% (1%) of the mean.²² The results are consistent with H2: cells see an increase in the likelihood of conflict if they are located along a river and the cells downstream, as opposed to upstream, experience a temperature shock. More broadly these results point to the river network as a key determinant of the spatial diffusion of climatic shocks onto conflict.

Actors involved in conflict In Table B2 we break down conflict incidence into different categories based on the involved actors. We find effects both when looking at battles (i.e. events involving organized groups like national armies, rebel groups, and local militias) and when looking at violence against civilians (i.e. events in which an organized group attacks civilians).

21. The difference between the two coefficients is statistically significant ($p = 0.03$).

22. Percentages of the mean in this section are computed against the outcome mean of the relevant estimation sample. Notice that the mean for the on-river up-down sample of Table 2 is 0.10, slightly larger than the mean for the full lateral sample of Table 1 that is 0.09.

Agricultural roots of conflict In Table 3 we show that the conflict we capture is rooted in competition between different groups over agricultural resources. We use the short descriptions that come with each ACLED event to classify the type of conflict. We first classify as agricultural conflict those events that pertain to land, water, crops, or livestock. We then further break down these conflicts between those that are related to land and water (factors of production) and those that are related to crops and livestock (outputs). We classify an event as agricultural conflict, factor conflict, or output conflict by matching the ACLED event notes against curated lists of keywords for each category, with agricultural conflict defined as the union of the factor and output categories.²³ The results show that, for both the lateral and the up-down design, the effects on agricultural conflict are larger relative to the mean than in the baseline specification. Moreover, we find strong effects for both factor and output conflict (McGuirk and Burke 2020). We notice that even with this breakdown between factor and output conflict our design cannot separate two channels: groups may temporarily move to contest water and the output it sustains, or they permanently migrate and generate conflict through the resulting pressure on land and resources. Both are consistent with our estimates, and we view them as complementary parts of the same mechanism rather than competing explanations.

Farmer-herder conflict Given the recent rise in violence in countries inhabited by pastoralists and the attention that farmers-herders violence has received in the literature (McGuirk and Nunn 2025; Eberle et al. 2025; Gehring and Schaudt 2026), we investigate whether water resources play a role in mediating this phenomenon. To this end, in Table 4 we differentiate between shocks originating in territories practicing agriculture and in territories traditionally inhabited by transhumant pastoralists.²⁴ Both types of economic activity depend on water and, when a shock erodes local access, the groups practicing them can seek it in nearby areas. Pastoralists can drive their cattle elsewhere in search of water and pasture, moving onto the land of neighboring communities. Farmers, being tied to fixed plots, respond differently: they may migrate to seize land closer to water, retaliate against upstream communities that over-extract from the shared river, or raid nearby water-rich areas to appropriate the output that water access has protected. In both cases, a temperature shock raises the value of contested water and can push groups to secure

23. Factor keywords include, among others, terms related to land disputes, grazing, and destroyed farmland, along with terms related to water disputes, access, wells, boreholes, and irrigation. Output keywords instead include terms related to cattle and livestock raiding, theft, and rustling, and to looted crops, harvests, or grain stores. Agricultural conflict corresponds to the union of the factor and output keyword matches. The complete keyword dictionaries and the classification procedure are reported in Appendix A.1 and Table A2.

24. We classify grid cells as agricultural if they contain agricultural land according to the PRIO-GRID dataset (Tollefson et al. 2012). To identify transhumant pastoralist territories, we combine the spatial distribution of ethnolinguistic groups with the *THP narrow* variable from the replication data provided by McGuirk and Nunn (2025). Importantly, these classifications are not mutually exclusive; a single grid cell may simultaneously host both agriculture and transhumant pastoralism.

it by moving onto, or preying upon, the territory of others.

In column 1 we find that higher temperatures in neighboring cells inhabited by transhumant pastoral groups increase conflict in water-rich cells. A similar pattern appears for agricultural neighbors (column 2). This suggests that farmers can either invade land in nearby water-rich areas or try to loot the agricultural output produced there. Both channels remain relevant when included jointly (column 3), and we do not find evidence that shocks in pastoral and agricultural neighbors differ significantly.²⁵

In columns 4-6 we focus on the up-down asymmetry along the river network. We observe that downstream temperature shocks in nearby *agricultural* territories have a large, precisely estimated effect on conflict (0.034 in columns 5-6). On the other hand, the impact of downstream shocks in nearby *pastoral* territories on conflict is small and imprecise (columns 4 and 6).²⁶

Taken together, the table suggests that the spatial distribution of water resources and the structure of the river network that connects them are key determinants of how climatic shocks propagate into violence across space. Our findings offer complementary evidence to McGuirk and Nunn (2025), who document how low rainfall pushes pastoralists to migrate earlier onto farmers' land, disrupting their traditional symbiotic relationship. We show that this pastoral-mobility channel is only one part of the picture: spatial inequality in water resources also shapes conflict dynamics originating in farmers' territories, and the propagation of shocks along the river network is driven entirely by shocks in territories using water for agriculture. Because the direction and reach of these spillovers are governed by the fixed, directional topology of the river network, our results are not simply characterizing cases in which herders invade farmers' territories in response to climatic shocks. They point instead to a novel mechanism in which the network itself channels where shocks travel: when hit by climatic shocks, farmers may invade other communities' land (whether inhabited by farmers or pastoralists), fight to prevent upstream over-extraction, or loot agricultural output (McGuirk and Burke 2020; Berman et al. 2021; Eberle et al. 2025). More broadly, this suggests that the spatial spillovers of climate shocks on farmer-herder conflict emphasized in the literature are, to some degree, shaped by hydrology: the river network determines not only whether shocks diffuse, but in which direction and toward whom.

5.1. Robustness and Sensitivity Checks

In this subsection, we provide evidence in support of our interpretation of the main results. We start by showing that results are robust to including a series of characteristics which may be correlated with both water richness and conflict. Then we perform a series of sensitivity checks,

25. A formal t -test fails to reject the null that the two coefficients are the same ($p = 0.69$).

26. The coefficients are essentially unaffected when both shocks are inserted at the same time, and a t -test rejects their equality ($p < 0.01$).

showing that the results are not driven by our specific choice of data or shock definition, or by specific countries or years.

Additional cell characteristics The first set of robustness checks focuses on adding control variables to address concerns on potential confounders. For the lateral exposure dynamics, we interact local shocks with the relevant characteristics. For the up-downstream channel, we include the relevant characteristics interacted with year fixed effects.

In Table B3 we interact lateral temperature with population density in the year 2000, share of the cell containing urban area, cell suitability for agriculture, presence of mines, and cell suitability for artisanal gold mining. The interaction between lateral exposure and water richness remains positive and significant throughout; its magnitude is stable across most controls but attenuates by roughly a third when we absorb agricultural suitability, the characteristic most closely tied to water richness. In Table B4 we interact the same characteristics with yearly time indicators; again, results are essentially unchanged. The final column of each table includes all these characteristics jointly: the lateral interaction remains positive and significant (attenuating from 0.006 to 0.004), and the up-down asymmetry likewise survives (the difference between downstream and upstream shocks remains significant at $p = 0.04$).

In Tables B5 and B6 we turn to factors linked to the geographic location of water resources. Rivers can constitute natural borders between countries and/or different ethnic groups, and borders may be correlated with higher risk of violence. Moreover, lakes and rivers may be closer or farther from large urban centers than the typical cell, and this can be correlated with institutional presence or market access. Finally, given that cells located on the coastline have no cells downstream by construction and that rivers tend to get larger volume as they flow toward the sea, we control for distance from the coast. Our estimates are largely unaffected by the inclusion of these controls; they are attenuated only when controlling for travel time to urban centers, but remain highly significant and economically meaningful. Including all border and remoteness measures jointly, in the final column of each table, leaves the lateral interaction positive and significant (0.003) and the up-down asymmetry significant at conventional levels ($p = 0.06$).

Ethnic grievances Recent literature stresses how climate shocks increase conflicts between different ethnic groups, and in particular among pastoralist and farmers groups. If rivers and lakes are more likely to be ethnic borders than the typical cell, then water-rich cells may be correlated with these ethnic grievances factors. In Tables B7 and B8 we check whether the impact of higher temperatures on conflict over water resources is robust to controlling for the presence of mixed settlements with both farmers and herders, ethnic polarization, and ethnic fractionalization.²⁷ The inclusion of these measures does not impact our main estimates of interest.

27. We take these measures from the replication data in Eberle et al. 2025.

Balance of upstream and downstream neighbors The up-down asymmetry is identified from the contrast between shocks originating upstream and downstream of a cell, so a natural concern is that downstream neighbors are systematically different places than upstream ones. In Table B9 we compare, for each on-river cell, the average characteristics of its upstream and downstream neighbors, and find them well balanced: population density, urbanization, agricultural suitability, market access, and ethnic composition do not differ significantly across the two groups. The only sizable differences are that downstream neighbors are closer to the coast and closer to ethnic borders. Overall, this balance reassures us that the observed asymmetry between upstream and downstream shocks is due to the direction of water flow rather than to some other characteristics being systematically different between upstream and downstream neighbors.

Competing climatic channels Weather is spatially correlated, so our exposure measures could pick up climatic variation other than heat shocks in the neighborhood. Table B10 addresses the two main candidates: own-cell heat and precipitation. Columns 1 to 3 report the lateral specification, columns 4 to 6 the up-downstream one. In column 1 we allow own-cell temperature to have a differential effect in water-rich cells. The lateral interaction of interest remains positive and of comparable magnitude; the own-cell interaction is itself small and insignificant. In column 4 we control for temperature in lateral cells: the up-down asymmetry is unchanged. Columns 2 and 5 control for the own-cell rainfall deficit.²⁸ We find that the rainfall coefficient is small and statistically indistinguishable from zero. Importantly, it leaves our coefficients of interest unchanged. Columns 3 and 6 include all climatic controls jointly, with no impact on the estimates.

Alternative shock definitions We also explore alternative options for measuring our main source of exogenous variation, namely climatic shocks. In Tables B11 and B12 we replicate our two main tables in full, using an alternative measure of extreme heat, the number of killing degree days (KDD) (Schlenker and Roberts 2009). The lateral results mirror the baseline: it is positive and highly significant, holds with time-varying discharge, and is driven by shocks originating in water-poorer neighbors. In the up-down specification, downstream shocks have a positive and significant effect, while the upstream coefficient is half the size and significant only at the ten percent level. Here the evidence is weaker: with KDD we fail to reject the null that the upstream and downstream shocks are the same ($p = 0.27$), though the point estimate for downstream shocks remains larger and more precisely estimated than the one for upstream shocks. Additionally, we explore the robustness of our effects to considering smaller or larger neighborhoods when defining exposure to shocks. In Table B13 we report results by considering several alternative radii, from 180 km (as in the seminal work by Harari and Ferrara 2018) to 400 km. Moreover, we show that results are similar if weight exposure by distance among cells, so that shocks in closer

28. We define rainfall deficit as minus the log of rainfall millimeters in a month, following the definition in Gehring and Schaudt 2026.

cells receive a larger weight. Results are essentially unaffected by these alternative definitions.

Lags In Table B14 we show that our results are robust to including lagged conflict incidence and lagged climatic shocks. The contemporaneous effects remain positive and significant throughout. Lagged shocks also carry some explanatory power: for the lateral interaction the lagged term is of similar magnitude to the contemporaneous one, while along the river network the contemporaneous downstream effect clearly dominates its lag. This is consistent with the effect of temperature shocks on conflict being concentrated in the same and immediately following year.

Water richness In Figure B5 we show the impact of lateral temperature shocks for different quintiles of water discharge. The results are larger for cells with greater amounts of water. We start to observe an impact for cells with discharge from the third quintile. However, the results are not driven only by large rivers: in Tables B15 and B16 we drop more and more cells with high discharge, until we only retain the ones below the sample median, and the results are essentially unaffected. This shows that our results are not driven by the largest rivers or lakes. Our conclusions are also robust to alternative operationalization of water richness and of the on-river sample. Table B17 shows that the lateral interaction remains positive and significant when we measure water richness with the plain log of discharge, $\log(\text{discharge})$, rather than the $\log(1 + \text{discharge})$ transformation used in our main specifications, or with a binary indicator for above-median discharge (columns 1 and 2). The downstream versus upstream asymmetry likewise persists under stricter definitions of the on-river sample based on seasonal-river length (columns 3 and 4).

Alternative conflict variables In Table B18 we check the robustness of our estimates using an alternative dependent variable, built from a different conflict dataset, UCDP-GED. Unlike ACLED, these data contain larger-scale violent events.²⁹ We report results for all conflicts (columns 1 and 2), for conflicts involving state forces as one of the actors (columns 3 and 4) and for conflicts not involving state actors (columns 5 and 6). For the lateral specification, we find that the interaction between lateral temperature and discharge is always positive, but it is statistically significant only for conflicts not involving state actors. For the on-river specification, across all categories of conflict, we still detect a large and precisely estimated effect for downstream shocks, while the effect of upstream shocks is smaller and not significant. The difference between the two coefficients is statistically significant for any conflict ($p = 0.03$) and for non-state conflict ($p = 0.02$), but not for conflict involving the state ($p = 0.16$). Taken together, these results are coherent with the dynamics of conflict we describe being mostly related to smaller-scale local violence rather than

29. To be included in the UCDP database, a conflict event must have at least one fatality and the pair of actors involved in the event (i.e., the conflict dyad) must have produced at least 25 fatalities in at least one calendar year throughout the series. Additionally, at least one of the actors involved in the event must be an “organized actor,” such as the state or a politically organized rebel group or militia.

to large-scale conflict involving the state or rebel groups.³⁰

Additional sensitivity We show that results are not driven by specific countries or years. In particular, we re-run our main analysis in different samples leaving out one year or one country at the time, for the lateral specification (Figure B1) and for on-river specification (Figures B2 and B3). We show that throughout all the leave-one-out estimations, we cannot reject that the main coefficients of interest (the interaction between lateral shocks and water richness, downstream shocks, upstream shocks) are equal to the estimates in the full sample analysis.

Alternative inference In Tables B19 and B20, we show that the statistical significance of the coefficients of interest is largely robust to allowing the spatial correlation in the Conley standard errors to have a larger spatial cutoff,³¹ or to use an alternative inference procedure where we use two-way cluster standard errors at the cell and country level, or at the cell and climate-zone year level. The up-down asymmetry is significant at the five percent level for spatial cutoffs up to 400 km and at the ten percent level for wider cutoffs; however, when we two-way cluster at the cell and country level we fail to reject at conventional levels the null that the upstream and the downstream shocks are the same. Finally, we assess the downstream–upstream asymmetry using a restricted permutation test. For each reference cell, we randomly reassign the upstream and downstream labels across its on-river neighbors while preserving the cell’s observed number of upstream and downstream neighbors. We then reconstruct the two temperature-exposure measures and re-estimate the baseline specification. The observed difference between the downstream and upstream effects lies beyond the entire distribution of relabeled estimates (Appendix Figure B4).³² The observed downstream–upstream asymmetry is therefore substantially larger than that generated by randomly reallocating directional labels within reference cells.

5.2. Factors exacerbating water-related conflict

In this section, we investigate the factors that exacerbate the risk of water-related conflict by leveraging the multidimensional nature of our data, specifically, its temporal and spatial variation, as well as the origins of conflict-inducing shocks. This approach enables us to identify the

30. The point estimates are smaller under UCDP-GED than under ACLED, but this largely reflects GED’s lower base rate, since GED records only larger-scale, higher-fatality events. Scaled by the outcome mean, the effect for non-state conflict is comparable across datasets.

31. The stability of the Conley standard errors across cutoffs is not an artifact of the kernel: the number of cell pairs entering the spatial window rises more than fivefold from the 200 km to the 500 km cutoff.

32. The test statistic is the difference between the downstream and upstream coefficients. Across 1,000 permutations, the reference distribution has a mean of approximately 0.004, compared with an observed coefficient difference of 0.017. The positive center of the permutation distribution reflects structural differences retained by the restricted design, including the observed upstream and downstream neighbor counts and the resulting differences in exposure variation. Nevertheless, none of the simulated coefficient differences is as large as the observed value, yielding a finite-simulation-corrected one-sided p-value of less than 0.001.

underlying mechanisms driving violence and provides guidance to policymakers regarding the regions most vulnerable to violence related to water resources.

Cooperation costs Water resources may be distributed unequally, yet different groups can still cooperate and manage them together. For instance, according to the hydraulic theory, the formation of early states was partly motivated by the necessity of institutions for large-scale irrigation projects (Allen et al. 2023). Moreover, a symbiotic system has often existed between farmers and herders, with herders migrating to farmers' land during dry seasons. This traditional arrangement, especially when farmers' land is situated near rivers, can be seen as a norm that enables efficient sharing of water resources among different groups. Cooperation over water resources can be facilitated by formal institutions, such as local and national governments, or informal institutions, such as traditional leaders. Hence, we may expect that the effects of climatic shocks on conflict are stronger when cooperation costs are higher, for instance when groups seeking access to water belong to different countries or to different ethnicities. In this section, we leverage differences in the origin of shocks to test whether the effects of climatic shocks on conflict are stronger when the costs of cooperation are higher.

First, we focus on cooperation costs due to differences in formal institutions. More specifically, we test whether differences in the effects of shocks are stronger if they originate in a different country. In Table 5, columns 1 and 3, we find that shocks originating in the same country as the reference cell have a similar impact to those originating in another country, in both the lateral and the up-downstream channel. This suggests that differences in formal institutions among nearby territories do not make it harder to cooperate over resources. At the same time, these results suggest that climatic shocks can spill into nearby countries, highlighting the potential for creating inter-state tensions beyond the forms of local violence we focus on.

Second, given the importance of traditional leaders in governing land and water, and the prevalence of inter-ethnic grievances across the continent, we focus on cooperation costs due to ethnic differences. In particular, we use data on the geographic distribution of linguistic groups to measure where different ethnic groups reside today.³³ In Table 5, columns 2 and 4, we test for differential effects of shocks originating in another ethnic group's territory.³⁴ In the lateral

33. We use the data of ethno-linguistic groups from the ethnologue (Giuliano and Nunn 2018).

34. Because this decomposition requires each cell to be assigned to a linguistic group, columns 2 and 4 are estimated on the subset of cells with a non-missing Ethnologue language: 8,687 of the 10,202 cells for the lateral channel (85% of the cell-year observations) and 8,093 of the 9,103 on-river cells for the up-down channel (89%). The cells without a language assignment are non-random, concentrated in sparsely populated arid margins outside the mapped ethno-linguistic areas. Reassuringly, both baseline results are essentially unchanged on the restricted sample. Re-estimating the lateral specification of column 2 of Table 1 on the language-coded cells leaves the interaction between lateral temperature and water richness positive and highly significant (0.005, against 0.006 in the full sample). Re-estimating the specification of column 3 of Table 2 yields, if anything, a stronger up-down asymmetry: a downstream effect of 0.025 (against 0.021 in the full sample), an upstream effect that remains small and insignificant, and a rejection of the equality of the two coefficients at least as strong ($p < 0.01$). The differential ethnic pattern we report is therefore not

channel (column 2), the interaction between temperature and water richness does not differ significantly depending on whether the shock originates in the same or a different ethnic group.³⁵ In the up-downstream channel (column 4), downstream temperature shocks originating in another group's territory increase conflict substantially, while same-group downstream shocks have no significant effect, and we reject that the two are equal. Indeed, among same-group neighbors we cannot reject that upstream and downstream shocks have the same effect, whereas we clearly reject it for other-group neighbors: the up-down asymmetry that characterizes our main results is driven by shocks that cross ethnic lines.

National institutions The results above suggest that heat shocks trigger violence where cooperation over water is hardest to sustain: informal arrangements between different ethnic groups appear fragile, while differences in formal institutions between neighboring territories play little role. A natural question is then whether strong national institutions can attenuate the risk of conflict when shocks hit.

There are three main dimensions along which national institutions may attenuate conflict risk. First, a state able to exert its monopoly of violence over the territory can prevent violence from erupting irrespective of the quality of its institutions. Second, an effective government may provide safety nets, build infrastructure, and enforce water governance cooperation, allowing local communities to cope with shocks. Third, an effective government need not deploy that capacity evenly. It may concentrate public services and political power in the urban areas where its political base lies and therefore not provide support to the rural populations who are mostly affected by water-related conflict. We explore how the effects of shocks vary with three country-level measures capturing these dimensions: state control over the territory and rural inclusion,³⁶ taken from the V-Dem project (Coppedge et al. 2023), and government effectiveness, taken from the World Bank Governance Indicators (Kaufmann et al. 2011). For each index, we assign to each cell the sample average of its country and compare countries above and below the median of the cells' sample distribution. Because our proxies for institutional quality exhibit variation only at the country level, they may be correlated with several unobserved national characteristics. The three measures are positively but only moderately correlated with one another across countries (pairwise correlations of 0.35 to 0.44), so while each plausibly captures a distinct facet of institutional quality,

an artifact of the sample restriction. Both re-estimations are reported in Appendix Table B21.

35. While the interaction with water richness is not significant for other-group shocks, it is marginally significant (10% level) for same-group shocks. For same-group shocks the baseline effect is sizable, negative and significant (-0.025); combined with the small positive interaction, this implies that the total same-group effect is negative in water-poor cells and turns positive only at very high levels of discharge, where in any case it is not statistically distinguishable from zero. On the other hand, the marginal effect on other-group shock is positive, albeit imprecisely estimated, across the whole discharge distribution.

36. Rural inclusion reverse-codes the V-Dem index of political exclusion by urban-rural location, so that a higher value denotes more equal treatment of rural and urban populations.

we do not over-interpret differences in the point estimates across them. Consequently, these results should be interpreted as suggestive rather than causal.

With this caveat in mind, the estimates in Table 6 reveal a consistent pattern across both channels. In the even columns, downstream shocks are attenuated by roughly two-thirds in countries with strong territorial control (column 2), high government effectiveness (column 4), or greater rural inclusion (column 6), and in none of the three can we still reject that upstream and downstream shocks have the same effect: the up-down asymmetry effectively disappears. In the odd columns, the interaction between lateral shocks and water richness is likewise negative and of similar magnitude under all three measures, again cutting the effect by roughly two-thirds.³⁷ Overall these findings suggest that stronger national institutions dampen the transmission of temperature shocks into conflict.

While the pattern is common to the three measures, each points to a different way in which the state can defuse conflict over water. Territorial control is consistent with the state's monopoly of violence raising the cost of resorting to fighting, whatever the underlying dispute. Government effectiveness suggests instead that the state reduces the need to fight, by providing safety nets and effective land and water governance. Rural inclusion indicates that what matters is not only the level of state capacity but also whether rural populations hold a claim on public services and political power.

Climate change An unexpected and short-term increase in temperature and in the value of water might increase violence, as it raises incentives for upstream farmers to over-extract or leads communities to invade or raid locations along the river, without necessarily prompting groups to invest in informal arrangements of cooperation over water governance. However, the impact of long-term water scarcity and prolonged shocks on cooperation is ambiguous, as they also increase the benefits derived from such cooperative arrangements. Despite the increased value of water due to long-term scarcity, effective informal institutions may be able to resolve disputes peacefully (Haseeb 2024). Similarly, local communities may invest in adaptation technologies, switch to heat-resistant crops, or migrate and move outside of agriculture; these changes should reduce the risk of conflict, as they reduce the demand for competing uses of water.

Table 7 examines how the relationship between climatic shocks and conflict evolves with long run changes in water availability and prolonged shocks. In particular, in columns 1 and 2 we interact the shocks with an indicator equal to one for cells whose 200 km neighborhood's total discharge is lower in the period 1997–2023 than in the pre-sample baseline (1979–1988). The triple interaction is positive and significant. Evaluating the lateral effect directly, the interaction between temperature and water richness is 0.008 (significant at the 1% level) where water has

37. The lateral interaction is statistically significant for territorial control (column 1) and rural inclusion (column 5), and slightly smaller and less precisely estimated for government effectiveness (column 3).

declined, against 0.002 and insignificant where it has not: the lateral effect is concentrated in areas of secular water decline. Column 2 shows a parallel pattern along the river network, where the effect of downstream shocks is larger in regions experiencing water decline, though the interaction itself is significant only at the 10% level. Columns 3-6 turn to persistence in *temperature* shocks. Temperature shocks are measured using rolling temperature, with two-year averages in columns 3-4 and four-year averages in columns 5-6. Relative to the contemporaneous estimates in Tables 1 and 2, the rolling-exposure coefficients are larger in magnitude. In the lateral specifications, the effect of temperature shocks in water-rich cells rises from 0.010 to 0.014 as the window lengthens; in the up-down specification, the effect of downstream shocks rises from 0.027 to 0.037, while upstream effects remain smaller and imprecise.

Taken together, we do not detect any attenuation of the effect under long-run water decline or prolonged shocks; if anything, both amplify it. The up-down asymmetry remains statistically significant under two- and four-year temperature averages ($p = 0.06$ in both).³⁸ We find no evidence of improved informal cooperation or of adaptation attenuating the effect under long-run water decline or prolonged shocks. This suggests that the conflict risk we investigated in this paper may escalate in the future, as climate change is projected to both increase water scarcity and lead to warmer years.

6 — Conclusion

This paper examines how the geography of water resources influences local violence across Africa from 1997 to 2023. We combine detailed data on hydrology, river network topology, and climatic shocks. The results show that water can generate a conflict resource curse. Because water is distributed unevenly across space, heat shocks increase violence in nearby water-rich areas. When a cell is exposed to a one degree increase in neighborhood temperature, the likelihood of conflict is 3.3 percentage points higher for a cell at the 75th percentile of water discharge than for one at the 25th. This effect corresponds to 36% of the mean incidence of conflict.

River networks also shape the spatial spillovers of climate shocks. The effect on violence is stronger for heat shocks that originate downstream. This pattern is consistent with upstream groups having greater ability to extract water before it flows downstream, giving them a nonviolent margin of adjustment when shocks occur. A one degree increase in downstream temperature

38. Because a multi-year moving average has mechanically lower variance than the annual shock, part of the larger rolling coefficient is expected even under an unchanged relationship. Rescaling by the standard deviation of each exposure measure, the effect of a one standard deviation increase changes little across horizons: for the lateral interaction it goes from 0.003 at the annual frequency to 0.004 at the four-year average, and for downstream shocks from 0.009 to 0.011. We therefore read these results as showing no attenuation under prolonged shocks, rather than as a precise measure of intensification.

raises the likelihood of conflict by 2.1 percentage points, or 21% of the mean. By contrast, the effect of upstream shocks is an order of magnitude smaller and not statistically significant.

The social and institutional environment conditions these effects. The effects are larger when shocks along the river originate in the territory of a different ethnic group. This suggests that cooperation over water is harder to sustain across ethnic lines. We also find suggestive evidence that the effects are smaller in countries with stronger and more inclusive national institutions. Geography and adverse shocks need not generate violence where the state has the capacity and willingness to sustain cooperation.

The results also speak to the long-run implications of climate change. The water conflict nexus is stronger when temperature increases persist over several years, consistent with limited adaptation to sustained warming. It is also stronger in areas where surface water availability has declined over the past forty years. As climate change raises temperatures and reshapes the distribution of freshwater across Africa, the risk of conflict over water may intensify.

Finally, our findings highlight the importance for governments and development agencies of considering how policies that affect water access may lead to increased violence. For instance, the design of irrigation and water infrastructure projects should take account of externalities on downstream communities. Similarly, conservation areas, large-scale land acquisitions, and commercial agriculture projects should also be assessed for whether they restrict access to water bodies on which local communities rely.

While this paper focuses on local violence, the same logic may extend to international relations. The ongoing geopolitical tensions surrounding the Grand Ethiopian Renaissance Dam provide a salient example (Climate Diplomacy 2023a). Future research should examine how river network structure and the control of water resources affect conflict and cooperation across borders.

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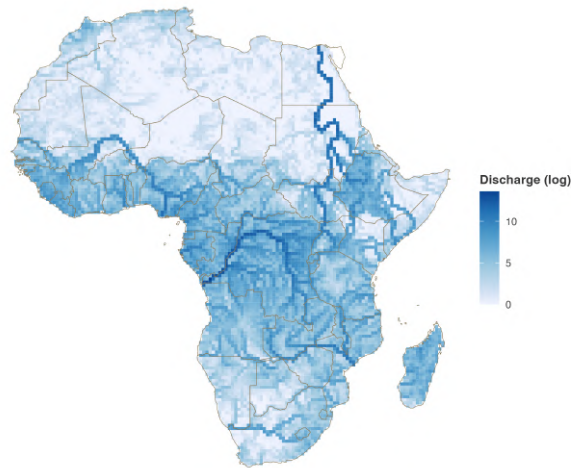
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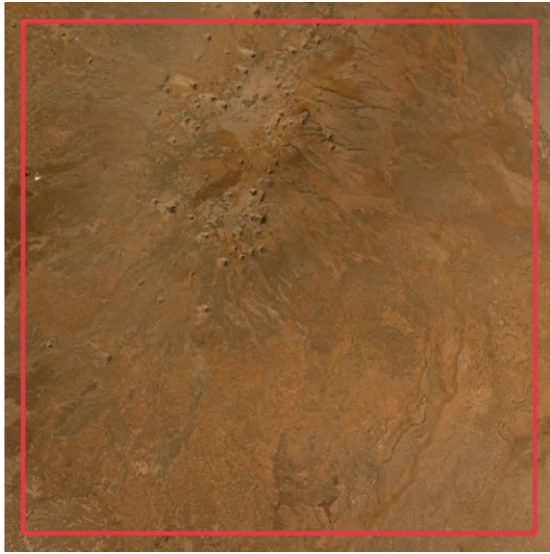
Figures and Tables

Figure 1 — Water richness



Notes: Cell-level (Log) average yearly discharge in m³/s over the sample period 1997-2023. Darker color indicates areas with higher average discharge. Water discharge data have been taken from Harrigan et al. 2020.

Figure 2 — Visualization of cells with low and high water discharge



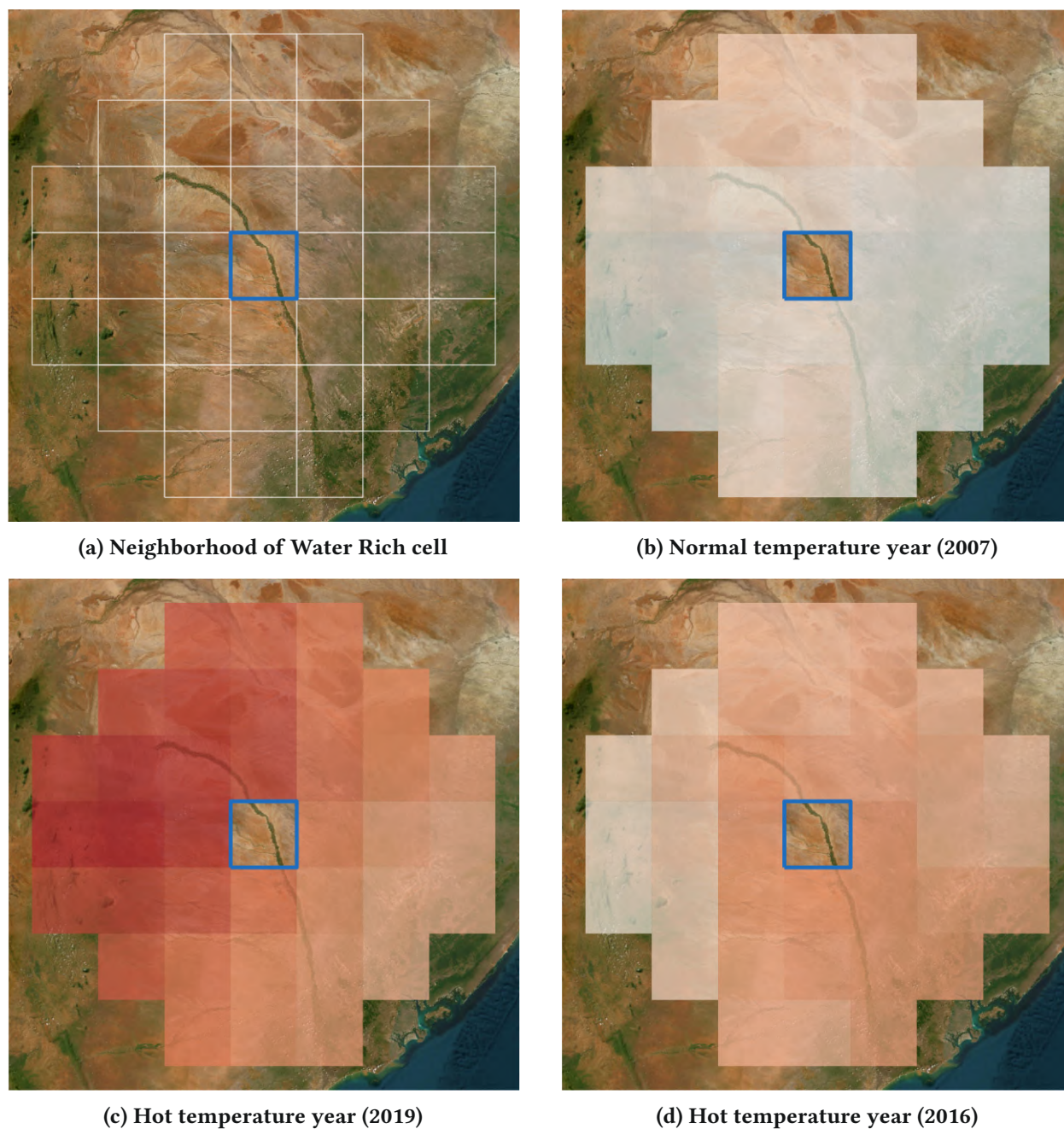
(a) Low Discharge (p3)



(b) High Discharge (p88)

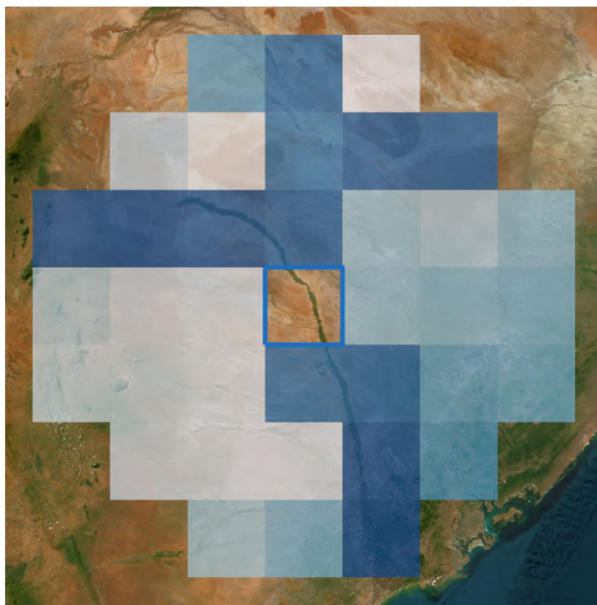
Notes. Satellite imagery (Esri WorldImagery) of two representative grid cells in Kenya. The left panel shows a cell at the 3rd percentile of the *Water Discharge* distribution; the right panel shows a cell at the 88th percentile.

Figure 3 — Visualization of neighborhood and exposure to temperature shocks

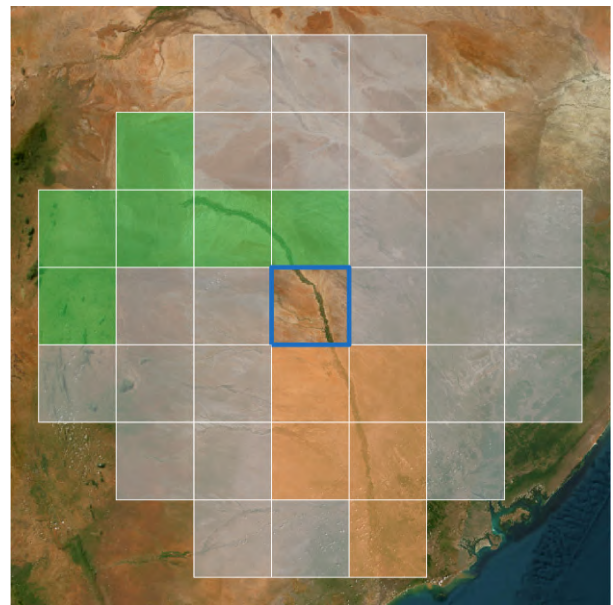


Notes. Top left: Satellite imagery (Esri WorldImagery) of a representative grid cells in Kenya and of its neighborhood (200 km radius). Top right and bottom: neighborhood temperature in different years. Darker shades of red indicate hotter temperature.

Figure 4 – Visualization of spatial inequality in water resources and river network connections



(a) Discharge (m3/s, log) of cells in neighborhood



(b) River network connections

Notes. The left panel shows the discharge of each cell in the neighborhood. Darker shades of blue indicate larger values of discharge (m3/s, log). The right panel shows the river network connections of each cell in the neighborhood. Green cells are upstream to the focal cells, orange cells are downstream, and gray cells are lateral. The focal cell is outlined in blue.

Table 1: Temperature shocks, spatial inequality in water resources, and conflict

	Any Conflict			
	(1)	(2)	(3)	(4)
Temp. own	-0.016 (0.012)	-0.018 (0.012)	-0.017 (0.012)	-0.022** (0.010)
Temp. lateral	0.017 (0.014)	-0.0001 (0.014)	0.002 (0.014)	
Temp. lateral × Discharge (log)		0.006*** (0.001)		
Discharge (log, annual)			0.001 (0.002)	
Temp. lateral × Discharge (log, annual)			0.006*** (0.001)	
Temp. lateral water-poor				-0.002 (0.005)
Temp. lateral water-rich				0.006 (0.011)
Temp. lateral water-poor × Discharge (log)				0.006*** (0.002)
Temp. lateral water-rich × Discharge (log)				0.0003 (0.002)
Cell FE	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.09	0.09	0.09
R ²	0.44	0.44	0.44	0.44
Cells	10,202	10,202	10,202	10,202
Observations	275,454	275,454	275,454	275,454
H ₀ : Water-poor = Water-rich origin (p)				0.05

Notes. The table reports estimates from Equation 1. The unit of observation is a $0.5^\circ \times 0.5^\circ$ grid cell and year. The dependent variable is any ACLED conflict incidence in a cell-year. *Temp. own* is cell-year temperature demeaned at the cell level; *Temp. lateral* is the mean demeaned temperature in neighboring lateral cells within 200 km, excluding the cell itself (Section 3.2). Column (2) interacts lateral exposure with *Discharge (log)* computed from the sample-period mean discharge (1997–2023). Column (3) instead interacts lateral exposure with *Discharge (log, annual)* computed from cell-year mean discharge ($\log(1 + \text{discharge}_{it})$). Column (4) splits lateral exposure into water-poor and water-rich neighbors, interacted with sample-period *Discharge (log)*. All columns include cell fixed effects and country-year fixed effects. Conley standard errors accounting for a spatial correlation of 200km and infinite serial correlation are reported in parenthesis. Statistical significance is represented by * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 2: Temperature shocks, river network, and conflict

	Any Conflict		
	(1)	(2)	(3)
Temp. own	-0.018** (0.008)	-0.005 (0.006)	-0.019** (0.008)
Temp. down	0.021*** (0.006)		0.021*** (0.006)
Temp. up		0.005 (0.004)	0.004 (0.004)
Cell FE	✓	✓	✓
Country-Year FE	✓	✓	✓
Dep. Var. Mean	0.10	0.10	0.10
R ²	0.44	0.44	0.44
Cells	9,103	9,103	9,103
Observations	245,781	245,781	245,781
H ₀ : Up = Down (p)			0.03

Notes. The table reports estimates from Equation 2. The unit of observation is a $0.5^\circ \times 0.5^\circ$ grid cell and year; the sample is restricted to on-river cells. The dependent variable is any ACLED conflict incidence in a cell-year. *Temp. own* is cell-year temperature demeaned at the cell level; *Temp. up* and *Temp. down* are the mean demeaned temperature in neighboring upstream and downstream cells within 200 km, excluding the cell itself (Section 3.2). Columns (1) and (2) include downstream and upstream exposure separately; column (3) includes both. The footer reports the p -value for the test that upstream and downstream exposure coefficients are equal. All columns include cell fixed effects and country-year fixed effects. Conley standard errors accounting for a spatial correlation of 200km and infinite serial correlation are reported in parenthesis. Statistical significance is represented by * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 3: Agricultural roots of conflict

	Agric. Conflict		Agric. Factor		Agric. Output	
	Lateral (1)	Up-Down (2)	Lateral (3)	Up-Down (4)	Lateral (5)	Up-Down (6)
Temp. own	-0.006 (0.008)	-0.006 (0.005)	-0.002 (0.007)	-0.002 (0.004)	-0.008 (0.007)	-0.007* (0.004)
Temp. lateral	-0.001 (0.009)		-0.003 (0.007)		0.002 (0.008)	
Temp. lateral $\times$ Discharge (log)	0.003*** (0.0008)		0.002*** (0.0006)		0.002** (0.0007)	
Temp. up		-0.002 (0.002)		-0.002 (0.002)		-0.002 (0.002)
Temp. down		0.010*** (0.003)		0.007** (0.003)		0.009*** (0.003)
Cell FE	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓
Dep. Var. Mean	0.02	0.02	0.01	0.02	0.01	0.01
R ²	0.30	0.30	0.25	0.25	0.26	0.26
Cells	10,202	9,103	10,202	9,103	10,202	9,103
Observations	275,454	245,781	275,454	245,781	275,454	245,781
H ₀ : Up = Down (p)		0.007		0.01		0.002

Notes. The table reports mechanism estimates for three agriculture- and land-water-related conflict-type outcomes. The unit of observation is a $0.5^\circ \times 0.5^\circ$ grid cell and year. Columns (1)–(2) use any agricultural conflict incidence as the outcome; columns (3)–(4) use land-water factor conflict incidence; columns (5)–(6) use agricultural output conflict incidence (all defined from event descriptions as in Section 3). Odd columns use the full sample and include lateral exposure interacted with *Discharge (log)* per Equation 1; even columns restrict to on-river cells and include upstream and downstream exposure with no discharge interaction, per Equation 2. *Temperature* is cell-year temperature demeaned at the cell level; lateral, upstream, and downstream exposures are defined in Section 3.2 (200 km). *Discharge (log)* is $\log(1 + \text{discharge})$ calculated as the mean cell discharge over 1997–2023 (Section 3.2). All columns include cell fixed effects and country-year fixed effects. Conley standard errors accounting for a spatial correlation of 200km and infinite serial correlation are reported in parenthesis. Statistical significance is represented by * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 4: Farmers-herders conflict

	Any Conflict					
	Lateral		Up-Down			
	(1)	(2)	(3)	(4)	(5)	(6)
Temp. own	-0.0004 (0.006)	-0.022*** (0.007)	-0.017** (0.008)	-0.003 (0.006)	-0.021*** (0.007)	-0.021*** (0.007)
Temp. lateral THP	-0.022*** (0.007)		-0.013* (0.007)			
Temp. lateral THP × Discharge (log)	0.007*** (0.002)		0.004** (0.002)			
Temp. lateral agri		0.015* (0.009)	0.018** (0.009)			
Temp. lateral agri × Discharge (log)		0.005*** (0.002)	0.003* (0.002)			
Temp. up THP				-0.003 (0.007)		-0.003 (0.006)
Temp. down THP				0.007 (0.007)		0.0009 (0.007)
Temp. up agri					0.005 (0.006)	0.006 (0.006)
Temp. down agri					0.034*** (0.008)	0.034*** (0.008)
Cell FE	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.09	0.09	0.10	0.10	0.10
R ²	0.44	0.44	0.44	0.44	0.44	0.44
Cells	10,202	10,202	10,202	9,103	9,103	9,103
Observations	275,454	275,454	275,454	245,781	245,781	245,781
H ₀ : THP = Agri, lateral × Discharge (p)			0.69			
H ₀ : Up = Down, THP (p)				0.31		0.70
H ₀ : Up = Down, Agri (p)					0.01	0.01
H ₀ : THP = Agri, Temp. down (p)						<0.01

Notes. The table reports mechanism estimates splitting neighbor temperature exposure by transhumant pastoralism (linguistic definition) and agricultural land use. The unit of observation is a $0.5^\circ \times 0.5^\circ$ grid cell and year. The dependent variable is any ACLED conflict incidence in a cell-year. Columns (1)–(3) use the full sample: column (1) includes lateral exposure from transhumant pastoral (THP narrow, linguistic) neighbors interacted with *Discharge (log)*; column (2) from agricultural neighbors; column (3) both. Columns (4)–(6) restrict to on-river cells and include upstream and downstream exposure from THP narrow (linguistic) neighbors (column 4), agricultural neighbors (column 5), or both (column 6), with no discharge interaction. Columns (1)–(2) and (4)–(5) report separate single-type specifications; the footer reports Wald tests that THP-narrow and agricultural neighbor coefficients are equal in column (3) for lateral exposure × *Discharge (log)* and in column (6) for downstream temperature exposure. Neighbors are classified as high or low on each trait using the sample median of the underlying cell-level index. *Temperature* is cell-year temperature demeaned at the cell level; exposures are defined in Section 3.2 (200 km). *Discharge (log)* is $\log(1 + \text{discharge})$ calculated as the mean cell discharge over 1997–2023 (Section 3.2). All columns include cell fixed effects and country-year fixed effects. Conley standard errors accounting for a spatial correlation of 200km and infinite serial correlation are reported in parenthesis; the same variance-covariance matrix is used for the footer Wald tests. Statistical significance is represented by * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 5: Cooperation costs and conflict

	Any Conflict			
	Lateral		Up-Down	
	Country (1)	Ethnic group (2)	Country (3)	Ethnic group (4)
Temp. own	-0.018* (0.010)	-0.004 (0.010)	-0.017** (0.007)	-0.017** (0.008)
Temp. lateral same neighbour	0.001 (0.012)	-0.025** (0.012)		
Temp. lateral other neighbour	-0.003 (0.006)	0.011 (0.007)		
Temp. lateral same neighbour $\times$ Discharge (log)	0.004** (0.002)	0.003* (0.002)		
Temp. lateral other neighbour $\times$ Discharge (log)	0.004** (0.002)	0.002 (0.002)		
Temp. up same neighbour			0.002 (0.004)	0.001 (0.004)
Temp. down same neighbour			0.013** (0.005)	0.004 (0.005)
Temp. up other neighbour			0.010 (0.008)	0.006 (0.005)
Temp. down other neighbour			0.016** (0.008)	0.025*** (0.007)
Cell FE	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.10	0.10	0.11
R ²	0.44	0.44	0.44	0.44
Cells	10,202	8,687	9,103	8,093
Observations	275,454	234,549	245,781	218,511
H ₀ : Same = Other neighbour, lateral $\times$ Discharge (p)	0.94	0.71		
H ₀ : Up = Down, same neighbour (p)			0.12	0.57
H ₀ : Up = Down, other neighbour (p)			0.63	0.03
H ₀ : Same = Other neighbour, Temp. down (p)			0.73	0.01

Notes. The table reports mechanism estimates decomposing neighbor temperature exposure by formal country borders and linguistic identity. The unit of observation is a $0.5^\circ \times 0.5^\circ$ grid cell and year. The dependent variable is any ACLED conflict incidence in a cell-year. Rows distinguish same vs. other neighbors; column sub-headers indicate whether exposure is split by country (columns 1 and 3) or linguistic group (columns 2 and 4). Columns (1)–(2) use the full sample and include lateral exposure interacted with *Discharge (log)* per Equation 1. Columns (3)–(4) restrict to on-river cells and include upstream and downstream exposure with no discharge interaction, per Equation 2. The footer reports four Wald tests: same-neighbor vs. other-neighbor equality for lateral exposure $\times$ *Discharge (log)* (columns 1–2); upstream vs. downstream equality for same-neighbor exposure and for other-neighbor exposure (columns 3–4); and same-neighbor vs. other-neighbor equality for downstream exposure (columns 3–4). *Temperature* is cell-year temperature demeaned at the cell level; lateral, upstream, and downstream exposures are defined in Section 3.2 (200 km). *Discharge (log)* is $\log(1 + \text{discharge})$ calculated as the mean cell discharge over 1997–2023 (Section 3.2). All columns include cell fixed effects and country-year fixed effects. Conley standard errors accounting for a spatial correlation of 200km and infinite serial correlation are reported in parenthesis. Statistical significance is represented by * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 6: State territorial control, national institutions, and conflict

	Any Conflict					
	State terr. control	Government effectiveness		Rural inclusion		
	(1)	(2)	(3)	(4)	(5)	(6)
Temp. own	-0.019 (0.012)	-0.018** (0.008)	-0.018 (0.012)	-0.020*** (0.007)	-0.018 (0.012)	-0.019*** (0.007)
Temp. lateral × Discharge (log)	0.009*** (0.002)		0.008*** (0.002)		0.008*** (0.002)	
Temp. lateral × Discharge (log) × High	-0.006** (0.003)		-0.004 (0.003)		-0.006** (0.003)	
Temp. up		-0.0008 (0.008)		-0.002 (0.009)		-0.006 (0.009)
Temp. down		0.033*** (0.009)		0.034*** (0.010)		0.037*** (0.010)
Temp. up × High		0.007 (0.009)		0.009 (0.009)		0.017* (0.010)
Temp. down × High		-0.021** (0.010)		-0.021** (0.010)		-0.027*** (0.010)
Cell FE	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.10	0.09	0.10	0.09	0.10
R ²	0.44	0.44	0.44	0.44	0.44	0.44
Cells	10,202	9,103	10,202	9,103	10,202	9,103
Observations	275,454	245,781	275,454	245,781	275,454	245,781
H ₀ : Marginal up = Marginal down, High state terr. (p)		0.45				
H ₀ : Marginal up = Marginal down, High gov. eff. (p)				0.41		
H ₀ : Marginal up = Marginal down, High rural incl. (p)						0.84

Notes. The table reports mechanism estimates interacting neighbor temperature exposure with above-median indicators of state territorial control and national institutions quality. The unit of observation is a $0.5^\circ \times 0.5^\circ$ grid cell and year. The dependent variable is any ACLED conflict incidence in a cell-year. Columns (1)–(2) use state control over territory (*High state terr.*), an index from the V-Dem project, columns (3)–(4) government effectiveness (*High gov. eff.*), from the World Bank Governance Indicators, and columns (5)–(6) rural inclusion (*High rural incl.*), the V-Dem index of political exclusion by urban–rural location reverse-coded so that *High* denotes greater inclusion of rural populations. Each indicator is a country’s 1997–2023 average assigned to every cell in that country; *High* equals one if that average is above the median computed across the sample’s cells, so the cutoff reflects the cell, not country, distribution, and is fixed over time. Odd columns use the full sample and augment Equation 1 with lateral exposure interacted with *Discharge (log)* and the institutions indicator; even columns restrict to on-river cells and augment Equation 2 with upstream and downstream exposure each interacted with the institutions indicator. All columns include cell fixed effects and country-year fixed effects. The footer reports the *p*-value for the Wald test that the marginal upstream and downstream exposure effects are equal in even columns, computed with Spatial-HAC standard errors. *Discharge (log)* is $\log(1 + \text{discharge})$ calculated as the mean cell discharge over 1997–2023. Conley standard errors accounting for a spatial correlation of 200km and infinite serial correlation are reported in parenthesis. Statistical significance is represented by * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 7: Long-run water decline and temperature increase, and conflict

	Water decline		Any Conflict			
	(1)	(2)	2-yr avg		4-yr avg	
			(3)	(4)	(5)	(6)
Temp. own	-0.018 (0.012)	-0.019** (0.008)	-0.023 (0.018)	-0.031*** (0.012)	-0.046* (0.026)	-0.049*** (0.017)
Temp. lateral	0.013 (0.014)		-0.015 (0.022)		-0.017 (0.033)	
Temp. lateral × Discharge (log)	0.002 (0.002)		0.010*** (0.002)		0.014*** (0.003)	
Temp. lateral × Water in region decline	-0.016*** (0.006)					
Discharge (log) × Temp. lateral × Water in region decline	0.006*** (0.002)					
Temp. up		0.009*** (0.003)		0.006 (0.006)		0.008 (0.008)
Temp. down		0.013** (0.005)		0.027*** (0.009)		0.037*** (0.013)
Temp. up x Water decline		-0.008 (0.006)				
Temp. down x Water decline		0.013* (0.007)				
Cell FE	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.10	0.09	0.10	0.10	0.11
R ²	0.44	0.44	0.45	0.44	0.46	0.46
Cells	10,202	9,103	10,202	9,103	10,202	9,103
Observations	275,454	245,781	265,252	236,678	244,848	218,472
H ₀ : Up = Down, at water decline = 1 (p)		0.02				
H ₀ : Up = Down, 2-yr avg (p)				0.06		
H ₀ : Up = Down, 4-yr avg (p)						0.06

Notes. The table reports mechanism estimates using long-run measures of water availability. The unit of observation is a $0.5^\circ \times 0.5^\circ$ grid cell and year. The dependent variable is any ACLED conflict incidence in a cell-year. Columns (1)–(2) interact neighbor temperature exposure with *Water in region decline*, an indicator equal to one when total discharge in the cell’s 200 km neighborhood is lower in the sample period (1997–2023) than in the pre-sample baseline (Section 3.2). Columns (3)–(4) replace the annual demeaned temperature shock with a two-period rolling average ($d = 1$); columns (5)–(6) use a four-period rolling average ($d = 3$). Within each panel, odd columns use the full sample and augment Equation 1 with lateral exposure interacted with *Discharge (log)*; even columns restrict to on-river cells and augment Equation 2 with upstream and downstream exposure. Columns (3)–(6) include the cell’s own rolling temperature shock. All columns include cell fixed effects and country-year fixed effects. The footer reports the p -value for the Wald test that upstream and downstream exposure coefficients are equal in even columns, computed with Spatial-HAC standard errors. *Discharge (log)* is $\log(1 + \text{discharge})$ calculated as the mean cell discharge over 1997–2023. Conley standard errors accounting for a spatial correlation of 200km and infinite serial correlation are reported in parenthesis. Statistical significance is represented by * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Online Appendix

Water Wars

Devis Decet Andrea Marcucci

A – Data Appendix

In this appendix we provide an overview of our data sources and the variables we construct from them (Section A.1 to Section A.6), after which we report summary statistics and spatial visualizations of the key variables in the study (Section A.7). Throughout, the unit of observation is a grid cell of $0.5^\circ \times 0.5^\circ$ (roughly 55×55 km at the equator) by year, and the estimation sample covers all of Africa over 1997–2023.

A.1. Conflict data

ACLED Our main source is the Armed Conflict Location & Event Data Project (ACLED), which records violent events across all African countries over 1997–2023 (Raleigh et al. 2010). ACLED gathers events from regional and national media, NGOs, and humanitarian organizations, and imposes no minimum fatality threshold, making it well suited to capture small-scale, localized violence. In the raw data the unit of observation is an event, with information on the actors involved, the event type, the location (longitude and latitude), fatalities, and the date. ACLED classifies events into event types, each divided into sub-event types.

We keep events with the highest geographic precision (ACLED geo-precision code 1 or 2), and assign each event to the grid cell with the closest centroid using its coordinates, aggregating to the cell-year level.³⁹ ACLED also provides a short free-text description of each event in the *notes* field. We use keyword dictionaries applied to this text to classify events as related to competition over agricultural factors of production (land and water) or over agricultural output (crops and livestock); see the measures below.

UCDP-GED For robustness we also use georeferenced events from the Uppsala Conflict Data Program (UCDP) Georeferenced Event Dataset (Sundberg and Melander 2013). To be included, an event must involve at least one fatality, and the dyad of actors must have caused at least 25 fatalities in a calendar year, with at least one organized actor (a state, rebel group, or militia). Relative to ACLED, UCDP is more focused on larger-scale conflict. We keep events with geo-precision 1 or 2 and assign each to the cell with the closest centroid.

39. Geo-precision code 1 is used when the source reports a specific town for which coordinates are available; code 2 when the source indicates a general area or the vicinity of a town.

Measures of conflict

- *Conflict incidence (ACLED)*: indicator equal to one if at least one *battle* or *violence-against-civilians* event (excluding the sub-event “Sexual violence”) is recorded in a cell-year. This is our main outcome.
- *Violence against civilians (ACLED)*: indicator equal to one if at least one violence-against-civilians event occurs in a cell-year, where an organized armed group inflicts violence on unarmed non-combatants. We exclude the sub-event type “Sexual violence”, as it is not related to the conflict dynamics we study.
- *Conflict incidence (UCDP-GED)*: indicator equal to one if at least one UCDP-GED event is recorded in a cell-year. Its state-based and non-state variants are used in robustness checks.
- *Agricultural factor conflict*: indicator equal to one if at least one conflict event in a cell-year has notes matching our land and water dictionaries. The dictionary captures competition over land and water as inputs (e.g. “land dispute”, “grazing”, “pasture”, “farmland”, crop destruction, “water dispute”, “access to water”, “borehole”, “watering point”, “irrigation water”).
- *Agricultural output conflict*: indicator equal to one if at least one conflict event in a cell-year has notes matching our output dictionary, which captures appropriation of surplus (e.g. “cattle raid”, “rustling”, “looted cattle”, “stolen livestock”, “harvest”).
- *Lagged conflict*: the one-year lag of *Conflict incidence (ACLED)*, used as a control in robustness checks.

The complete keyword dictionaries used to classify agricultural factor and agricultural output conflict are reported in Table A2. Figure A2 maps average conflict incidence by cell from ACLED and from UCDP-GED.

A.2. Hydrology data

Surface water We measure surface water using river discharge from the Global Flood Awareness System (GloFAS) (Harrigan et al. 2020). Discharge is the volume of water passing through a river cross-section per unit of time, in cubic meters per second, and accounts for all surface water bodies (lakes, ponds, rivers, streams). The data are produced by combining satellite observations, *in situ* measurements, and hydrological models, and are provided as daily values at a resolution of $0.05^\circ \times 0.05^\circ$. For each cell we sample discharge on the 15th day of each month and take the yearly average of these twelve monthly values. Our preferred, time-invariant measure of water richness is $\text{Discharge}(\log) = \log(1 + \overline{\text{Discharge}})$, where $\overline{\text{Discharge}}$ is the cell mean over 1997-2023 (Figure 1).⁴⁰ For robustness we also use a time-varying yearly measure.

40. We add one before taking logs because several cells have zero discharge.

Hydrographic basins and the river network We obtain river-network topology from the HydroBASINS dataset, part of the HydroSHEDS suite (Döll et al. 2003).⁴¹ HydroBASINS provides drainage-basin shapefiles at levels of increasing disaggregation; a basin is the area of land drained by a river and its branches. Following Strobl and Strobl (2011) and Eberle (2020), we use level-7 basins, whose average area is comparable to that of our grid cells. Because a square cell may intersect several basins, we assign each cell to the basin that contributes the most water: for each cell-basin intersection we compute the average discharge, and assign the cell to the basin whose intersection contains the highest water amount. Using the Pfafstetter coding system, we then construct, for every pair of cells within the same continental basin system, whether one is *upstream*, *downstream*, or *lateral* (neither) with respect to the other. Section A.6 describes this construction in detail.

We identify the presence of rivers from Messenger et al. (2021): a cell is classified as *on river* if its main basin contains a non-zero length of perennial or seasonal rivers and the cell has strictly positive average discharge over 1997–2023.

Finally, to capture long-run changes in water availability we construct *water in region decline*, an indicator equal to one if the total recent discharge in a cell’s 200 km neighborhood is below its pre-period (1979–1988) level, i.e. if the neighborhood has lost water over time.

A.3. Temperature data

Our treatment is a temperature (heat) shock. We use 2-meter air temperature from the ERA5 reanalysis of the Copernicus Climate Change Service (Hersbach et al. 2023), which combines worldwide observations with a model to produce gridded ($0.25^\circ \times 0.25^\circ$), hourly temperature. For each cell-year we aggregate the hourly data into mean temperature (T_{mean}). We then demean each cell’s temperature by its own 1997–2023 mean, so that the resulting variable *Temperature (demeaned)* captures deviations from the cell’s normal conditions rather than fixed cross-cell differences in climate.

Exposure to neighbors’ shocks For each cell we define a neighborhood as all cells whose centroid lies within 200 km.⁴² A cell’s exposure to temperature shocks is the average demeaned temperature of its neighbors, excluding the cell itself. Using the river-network coding, we split exposure into *lateral*, *upstream*, and *downstream* components (the average demeaned temperature of, respectively, lateral, upstream, and downstream neighbors), and we further split lateral exposure by whether the neighbor is more water-poor or more water-rich than the focal cell. As a robustness check we use killing-degree-days (KDD) as an alternative heat measure (Schlenker and Roberts 2009). This measure captures the number of fractional degree days in excess of a tem-

41. Available at <https://www.hydrosheds.org/>.

42. We explore the sensitivity of our results to alternative radii between 180 and 400 km.

perature threshold above which agricultural productivity suffers, demeaned and aggregated into neighborhood averages in the same way as the temperature shocks. We follow Hsiao et al. (2026) and set the temperature threshold to 32°C.

A.4. Rainfall and vegetation

Rainfall We use monthly precipitation from the ERA5 reanalysis of the Copernicus Climate Change Service (Hersbach et al. 2023), at 0.25° resolution. We aggregate to the cell-year level as the mean monthly precipitation in a cell. Our rainfall-based shock is the *rain deficit*, $-\log(\text{rainfall})$, so that higher values indicate drier conditions; like temperature, it is demeaned within cell.

Vegetation To validate that heat and water availability move agricultural output, we use Gross Dry Matter Productivity (GDMP) from the Copernicus Land Monitoring Service, a satellite measure of the growth rate of vegetation (dry biomass increase), aggregated from its native resolution to the cell-year level.

A.5. Other variables

- *Ethnic geography*: the geographic distribution of ethno-linguistic groups is originally from the *Ethnologue: Languages of the World* (Gordon Jr 2005) and we use the shapefile and the concordance mapping with the *Ethnographic Atlas* (Murdock 1967) from Giuliano and Nunn (2018). We link groups to their coding as transhumant pastoralists (*THP narrow*) from the replication data of McGuirk and Nunn (2025). We also create an indicator for cells intersecting a linguistic ethnic border.
- *Mixed Settlement and Ethnic*: measures of mixed settlement (i.e. cells inhabited both by farmers and pastoralist groups), ethnic polarization, and fractionalization are taken from the replication package of Eberle et al. (2025).
- *Population*: cell population in 2000 is from the Gridded Population of the World (Center for International Earth Science Information Network - CIESIN - Columbia University 2018).
- *Mining*: indicators for artisanal gold mining are from Girard et al. (2025).
- *Cell characteristics*: agricultural land cover, agricultural suitability, urban ground cover, and the presence of mines are from the PRIO-GRID (Tollefsen et al. 2012).
- *Travel time to urban center*: the mean travel time (log) from a cell to the nearest major city, from the PRIO-GRID (Tollefsen et al. 2012).
- *Distance to coast*: the geodesic distance (log, in km) from a cell centroid to the nearest African coastline, computed from the Natural Earth coastline.
- *National institutions*: we measure institutional quality with three time-invariant indicators. Each is a country's 1997–2023 average assigned to every cell in that country, and equal to one if that average is above the median computed across the sample's cells (so the cutoff reflects

the cell, not country, distribution). *State territorial control* captures the extent of recognition of the preeminent authority of the state over its territory and is an index from the Varieties of Democracy project (Coppedge et al. 2023). *Government effectiveness* measures the quality of public services and the government’s ability to formulate and implement sound policies, and is part of the Worldwide Governance Indicators (Kaufmann et al. 2011). *Rural inclusion* reverse-codes the V-Dem index of political exclusion by urban–rural location (Coppedge et al. 2023), so that a higher value denotes more equal treatment of rural and urban populations.

A.6. Upstream-downstream construction

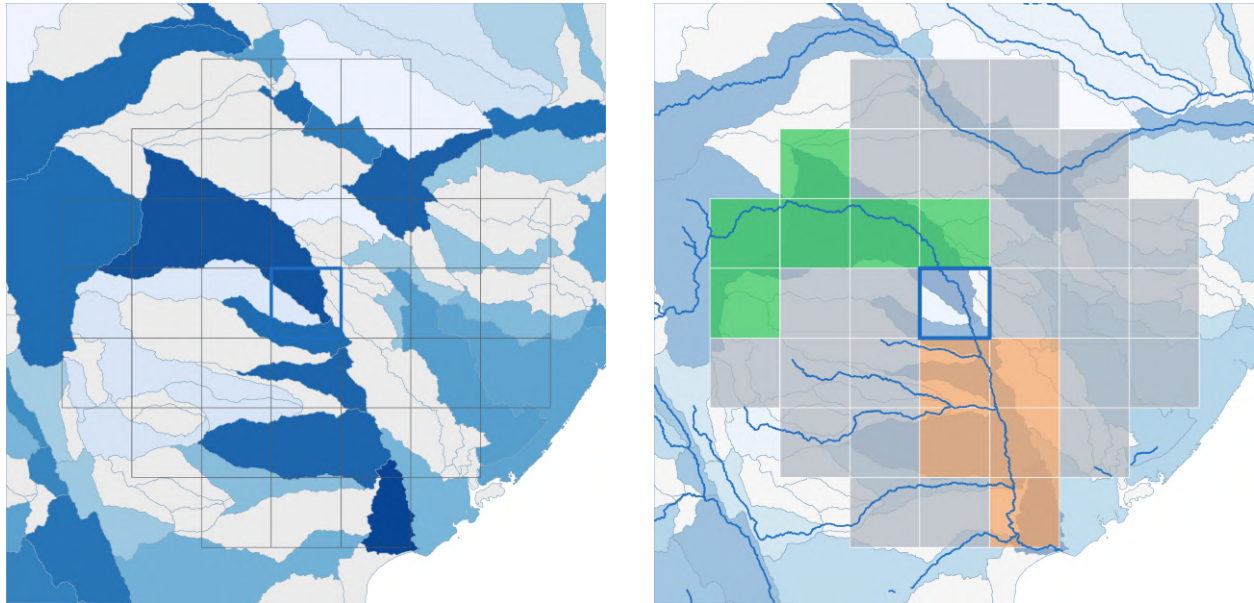
Section A.2 describes the basin partition, the level-7 disaggregation, and the rule assigning each cell to the basin that drains the largest amount of its water. Under the Pfafstetter coding system (Verdin and Verdin 1999), each basin carries information about its position along the river network, which is what lets us classify every pair of cells in the same continental basin system as upstream, downstream, or lateral.

We illustrate the methodology on the Tana river in Kenya, using the same water-rich cell used to visualize the temperature neighborhood elsewhere in the paper. In the left panel of Figure A1 we overlay the neighborhood of all cells whose centroid lies within 200 km of the blue reference cell at the center, together with the river basins in the area, shaded by their average discharge. In the right panel we display the resulting assignment over the same discharge-shaded basins: gray cells are within 200 km of the reference cell but have no upstream-downstream relationship with it; green (orange) cells are upstream (downstream); the blue lines are the rivers with the highest water presence (here, the Tana), which can be seen passing through the basins into and out of the reference cell.

A.7. Data visualization and summary statistics

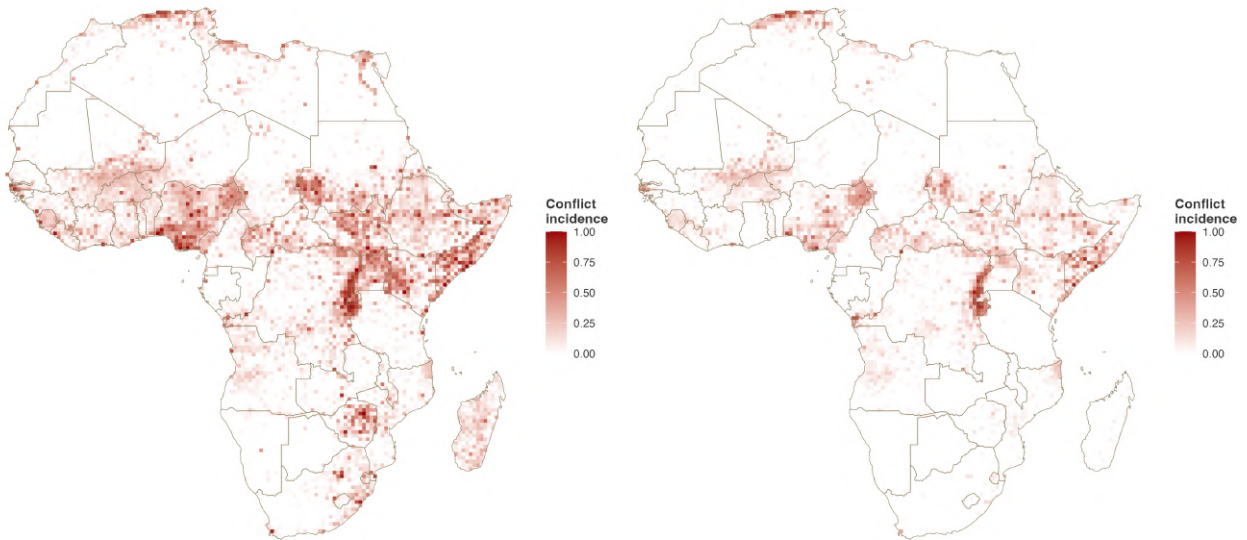
Table A1 reports summary statistics for the variables used in our main specifications, and Table A2 the keyword dictionaries used to classify agricultural conflict. Figures A2–A3 map the key variables across the continent: conflict incidence from ACLED and UCDP-GED, the agricultural and pastoralist (THP) classification of cells, and which cells lie on a river. Finally, Figure A4 illustrates the identifying variation behind hypothesis H1: after removing cell and country-year fixed effects, lateral temperature exposure still varies substantially across space and time, and this variation interacts with the fixed cross-sectional distribution of water richness.

Figure A1 – Tana river upstream and downstream relationship



Notes: The figure illustrates, using the Tana river in Kenya, how we build the upstream-downstream relationships. In the left panel we superimpose the grid for the neighborhood (cells within a 200 km radius) of the blue reference cell at the center with the river-basin shapefile, colored according to the average water discharge in each basin. In the right panel we show the resulting upstream-downstream relationships between cells over the same discharge-shaded basins. Green cells are upstream and orange cells downstream within the neighborhood of the main (blue) cell; gray cells are coded as neither upstream nor downstream, and the blue lines trace the rivers with the highest average discharge as they pass through the basins.

Figure A2 – Conflict incidence by cell: ACLED and UCDP-GED, 1997–2023



Notes. Spatial distribution of conflict incidence, 1997–2023: ACLED (left) and UCDP-GED (right). Darker cells have a higher share of years with at least one conflict event (Section A.1). Country borders are shown; cells outside the sample are shaded grey.

Table A1: Summary statistics: main variables

Variable	N	Mean	SD	Min	p25	Median	p75	Max
Conflict incidence (ACLED)	275,454	0.09	0.287	0	0	0	0	1
Battles (ACLED)	275,454	0.06	0.237	0	0	0	0	1
Violence against civilians (ACLED)	275,454	0.065	0.246	0	0	0	0	1
Agricultural factor conflict (land & water)	275,454	0.016	0.127	0	0	0	0	1
Agricultural output conflict	275,454	0.014	0.117	0	0	0	0	1
Temperature, demeaned	275,454	0	0.487	-1.906	-0.316	-0.036	0.291	2.339
Lateral exposure	275,454	0.113	0.471	-1.699	-0.194	0.074	0.391	2.262
Upstream exposure	275,454	0.07	0.388	-1.853	-0.036	0	0.204	2.4
Downstream exposure	275,454	0.09	0.437	-1.759	-0.138	0	0.321	2.47
Lateral exposure, water-poor neighbors	275,454	0.102	0.453	-1.823	-0.174	0.032	0.359	2.242
Lateral exposure, water-rich neighbors	275,454	0.105	0.458	-1.84	-0.168	0.028	0.362	2.377
Lateral exposure, pastoralist (THP) neighbors	275,454	0.079	0.4	-1.706	0	0	0.135	2.463
Lateral exposure, agricultural neighbors	275,454	0.07	0.383	-1.77	-0.08	0	0.251	2.338
Lateral exposure, same-country neighbors	275,454	0.114	0.476	-1.71	-0.198	0.075	0.398	2.323
Lateral exposure, other-country neighbors	275,454	0.063	0.372	-1.762	-0.006	0	0.157	2.449
Lateral exposure, same-language neighbors	275,454	0.087	0.426	-1.882	-0.121	0	0.296	2.386
Lateral exposure, other-language neighbors	275,454	0.081	0.402	-1.882	-0.095	0	0.274	2.394
Discharge (log)	275,454	3.863	3.099	0	0.965	3.51	6.31	13.58
On river (indicator)	275,454	0.892	0.31	0	1	1	1	1
High state territorial control (V-Dem)	275,454	0.499	0.5	0	0	0	1	1
High government effectiveness	275,454	0.499	0.5	0	0	0	1	1
High rural inclusion (V-Dem)	275,454	0.523	0.499	0	0	1	1	1
Agricultural land cover	275,454	15.923	24.479	0	0	2.383	23.028	99.917
Pastoralist cell (THP, narrow)	275,454	0.277	0.447	0	0	0	1	1

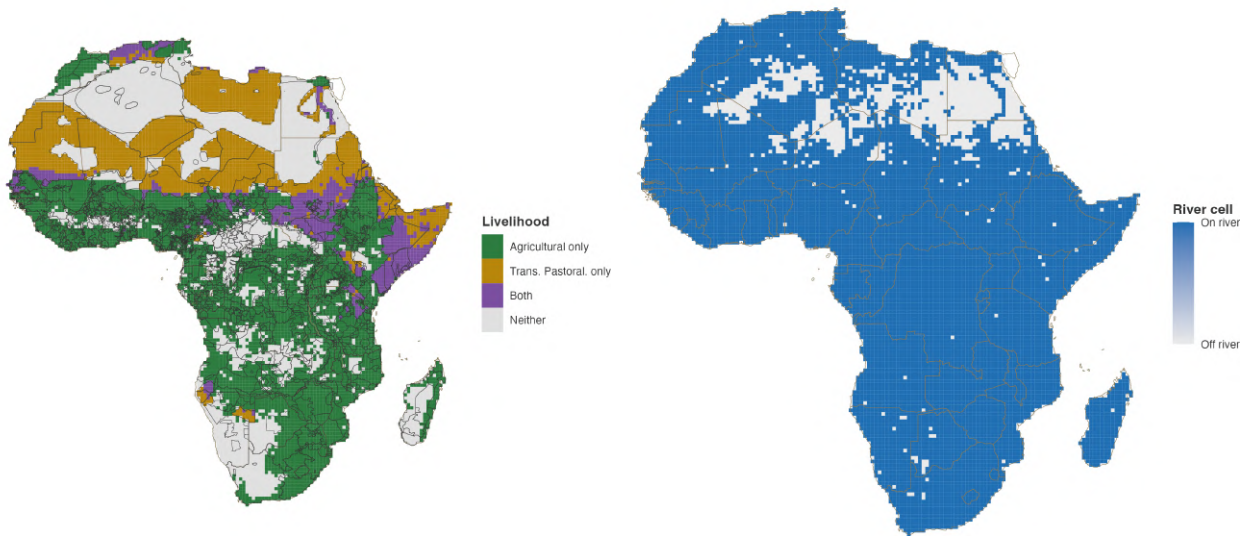
Notes. The table reports summary statistics for the variables used in the main specifications, over the full sample of grid cell-years, 1997–2023. Rows are grouped by domain (conflict outcomes; own-cell shock; neighborhood temperature exposures; exposures by neighbor type; water richness; national institutions; cell characteristics). Variable definitions and sources are given in Sections A.1–A.5. *Temperature, demeaned* and the temperature exposures are demeaned within cell; *Discharge (log)* is $\log(1 + \text{Discharge})$ using the cell mean over the sample period; the institutions indicators are median-split country averages.

Table A2: Keyword dictionaries for agricultural conflict classification

Category	Keywords
Agricultural factor (land and water)	“land dispute”; “dispute over land”; “control of land”; “over land”; “clash over land”; “land grab”; “farm land”; “land invaders”; “land battle”; “over cattle and land”; “invade land”; “over disputed land”; “over a piece of land”; “herd”; “pastoral”; “grazing”; “pasture”; “destroy crop”; “destroying crop”; “burned crop”; “burning crop”; “burned crops”; “destroyed crops”; “crops destroyed”; “crops burned”; “farmland”; “cropland”; “crop field”; “water dispute”; “dispute over water”; “over water”; “control of water”; “water rights”; “access to water”; “water access”; “water point”; “waterpoint”; “watering point”; “water hole”; “waterhole”; “watering hole”; “water well”; “borehole”; “bore hole”; “water pump”; “water source”; “water dam”; “water reservoir”; “irrigation canal”; “irrigation water”; “water channel”; “water canal”; “river access”; “in search of water”; “livestock watering”; “water his field”; “pasture and water”; “water and pasture”; “ownership of water”; “water resources”; “over a stream”; “farm irrigation”
Agricultural output	“cattle raid”; “cattle raiding”; “cattle theft”; “cattle rustl”; “livestock raid”; “livestock theft”; “livestock rustl”; “steal cattle”; “stealing cattle”; “stolen cattle”; “stolen animal”; “rustling”; “rustlers”; “raided cattle”; “cattle camp”; “looted cattle”; “stole cattle”; “seized livestock”; “seized cattle”; “stolen cows”; “looted cows”; “stole cows”; “loot cattle”; “loot livestock”; “loot cows”; “looting cattle”; “looting livestock”; “looting cows”; “cow”; “livestock”; “cattle”; “steal crop”; “stealing crop”; “food store”; “grain store”; “harvest”; “seized harvest”; “loot village”

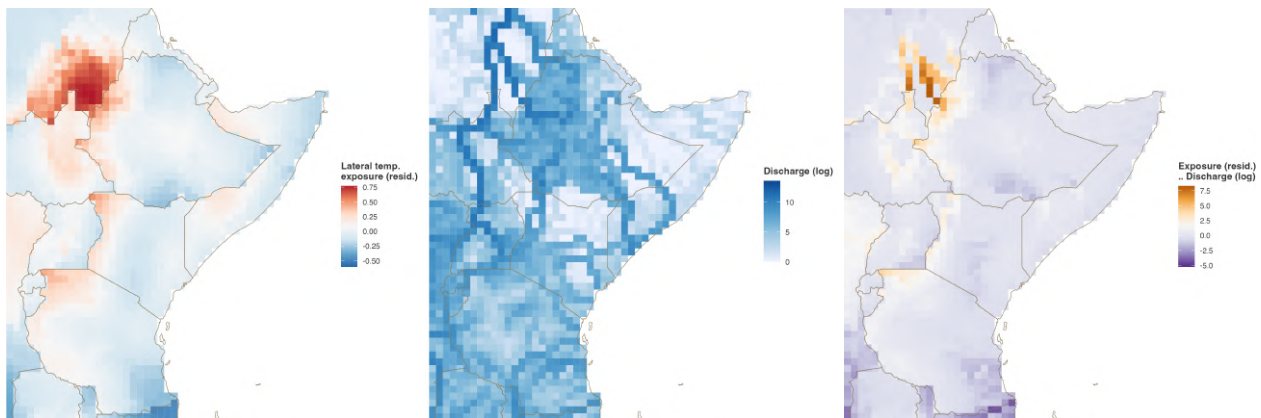
Notes. The table reports the complete keyword lists applied to the ACLED event *notes* field to classify agricultural conflict. An event is coded as agricultural *factor* conflict if its notes match the agricultural-factor dictionary (competition over land and water as inputs), and as agricultural *output* conflict if they match the output dictionary (appropriation of crops and livestock). See Section A.1.

Figure A3 — Economic livelihoods and on-river cells



Notes. Left: cell livelihood classification. A cell is *agricultural* if its share of agricultural land cover (PRIO-GRID) is above the sample median, and *pastoralist* if its intensity of transhumant pastoralism (THP narrow; McGuirk and Nunn 2025) is above the sample median. As the two are separate median splits, cells are shown as agricultural only (green), pastoralist only (ochre), both (purple), or neither (grey). Right: cells classified as on a river (positive perennial or seasonal river length in the main basin and positive average discharge). Water richness (Discharge (log)) is mapped in Figure 1. Country borders are shown; cells outside the sample are shaded grey.

Figure A4 – Sources of identifying variation (East Africa, 2016)



Notes. The figure illustrates the variation identifying hypothesis H1 in East Africa in 2016. The left panel maps lateral temperature exposure after residualizing on cell and country-year fixed effects, showing that identifying variation in neighbors' heat remains after the fixed effects. The middle panel maps water richness, Discharge (log), which is fixed over time. The right panel maps their interaction (residualized lateral exposure $\times$ water richness), the object H1 is identified from: conflict is predicted to rise most in water-rich cells whose lateral neighbors are hit by heat.

B — Additional Results

Notes common to all tables in this appendix. The unit of observation is a $0.5^\circ \times 0.5^\circ$ grid cell and year, 1997–2023. Unless a table states otherwise: the dependent variable is any ACLED conflict incidence in a cell-year; *Temperature* (*Temp. own*) is cell-year temperature demeaned at the cell level; *Exposure Lateral*, *Exposure Up* and *Exposure Down* (*Temp. lateral*, *Temp. up*, *Temp. down*) are the mean demeaned temperature of, respectively, the lateral, upstream and downstream neighbors within 200 km, excluding the cell itself (Section 3.2); *Discharge (log)* is $\log(1 + \text{discharge})$ calculated as the mean cell discharge over 1997–2023; specifications based on Equation 1 use the full sample and those based on Equation 2 are restricted to on-river cells; all columns include cell fixed effects and country-year fixed effects; and Conley standard errors accounting for a spatial correlation of 200 km and infinite serial correlation at the cell level are reported in parentheses. The significance legend is repeated in each table.

Table B1: Surface water and the effects of temperature on vegetation

	log(gdmp)			
	(1)	(2)	(3)	(4)
Temp. own	-0.166*** (0.031)	-0.308*** (0.042)	-0.270*** (0.043)	-0.259*** (0.043)
Discharge (log)	0.184*** (0.021)	0.183*** (0.021)		
Temp. own $\times$ Discharge (log)		0.041*** (0.007)	0.044*** (0.007)	0.044*** (0.007)
Rain deficit				-0.058** (0.025)
Country-Year FE	✓	✓	✓	✓
Cell FE			✓	✓
Dep. Var. Mean	2.1	2.1	2.1	2.1
R ²	0.75	0.75	0.92	0.92
Cells			10,199	10,199
Observations	249,013	249,013	249,013	248,920

Notes. The table reports OLS estimates of the relationship between vegetation productivity and temperature, water discharge, and rainfall. The dependent variable is log(GDMP) (gross dry matter productivity, a phytomass proxy) from ERA5 (Hersbach et al. 2023). *Rain deficit* is $-\log(\text{rainfall})$ so that higher values indicate less rainfall. Columns (1)–(2) include country-year fixed effects only; columns (3)–(4) add cell fixed effects. The sample is 249,013 cell-years with non-missing vegetation data. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B2: Type of conflict

	Battle		Violence vs. Civilians	
	Lateral (1)	Up-Down (2)	Lateral (3)	Up-Down (4)
Temp. own	-0.013 (0.010)	-0.016*** (0.006)	-0.012 (0.011)	-0.010 (0.007)
Temp. lateral	-0.0005 (0.011)		-0.001 (0.012)	
Temp. lateral × Discharge (log)	0.004*** (0.001)		0.006*** (0.001)	
Temp. up		0.0007 (0.003)		0.0009 (0.004)
Temp. down		0.017*** (0.005)		0.016*** (0.005)
Cell FE	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓
Dep. Var. Mean	0.06	0.07	0.07	0.07
R ²	0.38	0.38	0.41	0.41
Cells	10,202	9,103	10,202	9,103
Observations	275,454	245,781	275,454	245,781
H ₀ : Up = Down (p)		0.007		0.04

Notes. The table reports mechanism estimates for two ACLED conflict-type outcomes. Columns (1)–(2) use battle incidence as the outcome; columns (3)–(4) use violence-against-civilians incidence (defined from event descriptions as in Section 3). Columns (1) and (3) use the full sample and include lateral exposure interacted with *Discharge (log)* per Equation 1; columns (2) and (4) restrict to on-river cells and include upstream and downstream exposure with no discharge interaction, per Equation 2. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B3: Robustness: lateral temperature exposure interacted with cell characteristics

	Any Conflict					
	(1)	(2)	(3)	(4)	(5)	(6)
Temp. own	-0.018 (0.012)	-0.018 (0.012)	-0.021* (0.012)	-0.018 (0.012)	-0.017 (0.012)	-0.020* (0.012)
Temp. lateral	0.0008 (0.014)	-0.0005 (0.014)	-0.001 (0.014)	-0.0006 (0.014)	-0.010 (0.013)	-0.011 (0.013)
Temp. lateral × Discharge (log)	0.005*** (0.001)	0.006*** (0.001)	0.004*** (0.001)	0.006*** (0.001)	0.007*** (0.001)	0.004*** (0.001)
Temp. lateral × Population density (2000)	0.109** (0.043)					0.136** (0.056)
Temp. lateral × Urban share		0.004 (0.003)				-0.0008 (0.003)
Temp. lateral × Pixel suitability			0.081*** (0.015)			0.087*** (0.014)
Temp. lateral × Mines				0.021** (0.009)		0.014 (0.009)
Temp. lateral × Artisanal gold suitability					0.030*** (0.011)	0.025** (0.011)
Cell FE	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.09	0.09	0.09	0.09	0.09
R ²	0.44	0.44	0.44	0.44	0.44	0.44
Cells	10,202	10,202	10,202	10,091	10,065	9,956
Observations	275,454	275,454	275,454	272,457	271,755	268,812

Notes. The table reports robustness estimates from Equation 1. Each column adds *Exposure Lateral* interacted with a different cell characteristic on top of *Exposure Lateral* × *Discharge (log)*: column (1) population density (2000); (2) urban share; (3) pixel suitability; (4) mines; (5) artisanal gold suitability (from Girard et al. 2025). The final column includes all five interactions simultaneously. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B4: Robustness: upstream and downstream temperature exposure controlling for cell characteristics

	Any Conflict					
	(1)	(2)	(3)	(4)	(5)	(6)
Temp. own	-0.021*** (0.008)	-0.020*** (0.008)	-0.019*** (0.007)	-0.020*** (0.008)	-0.020*** (0.007)	-0.021*** (0.007)
Temp. up	0.004 (0.004)	0.004 (0.004)	0.004 (0.004)	0.004 (0.004)	0.004 (0.004)	0.003 (0.004)
Temp. down	0.021*** (0.006)	0.021*** (0.006)	0.019*** (0.006)	0.022*** (0.006)	0.020*** (0.006)	0.019*** (0.006)
Cell FE	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Population density (2000) × Year FE	✓					✓
Urban share × Year FE		✓				✓
Pixel suitability × Year FE			✓			✓
Mines × Year FE				✓		✓
Artisanal gold suitability × Year FE					✓	✓
Dep. Var. Mean	0.10	0.10	0.10	0.10	0.10	0.10
R ²	0.44	0.44	0.44	0.44	0.44	0.45
Cells	9,103	9,103	9,103	8,995	8,968	8,861
Observations	245,781	245,781	245,781	242,865	242,136	239,247
H ₀ : Up = Down (p)	0.02	0.03	0.03	0.03	0.04	0.04

Notes. The table reports robustness estimates from Equation 2. Each column adds year fixed effects with slopes on a different cell characteristic: column (1) population density (2000); (2) urban share; (3) pixel suitability; (4) mines; (5) artisanal gold suitability (from Girard et al. 2025). The final column includes all five slopes simultaneously. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B5: Robustness: lateral temperature exposure interacted with border proximity and remoteness

	Any Conflict						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Temp. own	-0.017 (0.013)	-0.018 (0.012)	-0.017 (0.012)	-0.018 (0.012)	-0.022* (0.012)	-0.017 (0.012)	-0.017 (0.012)
Temp. lateral	0.027 (0.020)	-0.003 (0.015)	-0.0009 (0.014)	0.002 (0.014)	0.172*** (0.033)	-0.013 (0.035)	0.151*** (0.050)
Temp. lateral × Discharge (log)	0.006*** (0.001)	0.006*** (0.001)	0.006*** (0.001)	0.006*** (0.001)	0.004*** (0.001)	0.006*** (0.001)	0.003*** (0.001)
Temp. lateral × Distance to country border (log)	-0.006** (0.003)						-0.006** (0.003)
Temp. lateral × Distance to linguistic border (log)		0.0009 (0.001)					-0.004 (0.002)
Temp. lateral × Country border			0.015 (0.013)				0.030** (0.014)
Temp. lateral × Linguistic border				-0.009* (0.005)			-0.022** (0.010)
Temp. lateral × Travel time to urban center (log)					-0.025*** (0.005)		-0.030*** (0.005)
Temp. lateral × Distance to coast (log)						0.002 (0.005)	0.015** (0.006)
Cell FE	✓	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.09	0.09	0.09	0.09	0.09	0.09
R ²	0.44	0.44	0.44	0.44	0.44	0.44	0.44
Cells	10,004	10,202	10,202	10,202	10,202	10,202	10,004
Observations	270,108	275,454	275,454	275,454	275,454	275,454	270,108

Notes. The table reports robustness estimates from Equation 1. Each column adds *Exposure Lateral* interacted with a different border or remoteness measure on top of *Exposure Lateral × Discharge (log)*: column (1) distance to country border (log); (2) distance to linguistic border (log); (3) country border cell; (4) linguistic border cell; (5) travel time to urban center (log); (6) distance to coast (log). The final column includes all six interactions simultaneously. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B6: Robustness: upstream and downstream temperature exposure controlling for border and remoteness characteristics

	Any Conflict						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Temp. own	-0.018** (0.008)	-0.019** (0.008)	-0.020** (0.008)	-0.020** (0.008)	-0.012* (0.007)	-0.020*** (0.008)	-0.011 (0.007)
Temp. up	0.003 (0.004)	0.004 (0.004)	0.003 (0.004)	0.004 (0.004)	0.001 (0.004)	0.004 (0.004)	-0.0001 (0.004)
Temp. down	0.021*** (0.006)	0.021*** (0.006)	0.022*** (0.006)	0.021*** (0.006)	0.017*** (0.006)	0.021*** (0.006)	0.013** (0.005)
Cell FE	✓	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓
Distance to country border (log) × Year FE	✓						✓
Distance to linguistic border (log) × Year FE		✓					✓
Country border × Year FE			✓				✓
Linguistic border × Year FE				✓			✓
Travel time to urban center (log) × Year FE					✓		✓
Distance to coast (log) × Year FE						✓	✓
Dep. Var. Mean	0.10	0.10	0.10	0.10	0.10	0.10	0.10
R ²	0.44	0.44	0.44	0.44	0.44	0.44	0.45
Cells	8,907	9,103	9,103	9,103	9,103	9,103	8,907
Observations	240,489	245,781	245,781	245,781	245,781	245,781	240,489
H ₀ : Up = Down (p)	0.02	0.03	0.01	0.03	0.05	0.02	0.06

Notes. The table reports robustness estimates from Equation 2. Each column adds year fixed effects with slopes on a different border or remoteness measure, in the same order as Table B5: column (1) distance to country border (log); (2) distance to linguistic border (log); (3) country border cell; (4) linguistic border cell; (5) travel time to urban center (log); (6) distance to coast (log). The final column includes all six slopes simultaneously. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B7: Robustness: lateral temperature exposure interacted with ethnic composition

	Any Conflict		
	(1)	(2)	(3)
Temp. own	-0.017 (0.013)	-0.017 (0.013)	-0.016 (0.013)
Temp. lateral	-0.0009 (0.015)	0.004 (0.016)	0.007 (0.016)
Temp. lateral $\times$ Discharge (log)	0.006*** (0.001)	0.006*** (0.001)	0.006*** (0.001)
Temp. lateral $\times$ Mixed settlements	0.007 (0.011)		
Temp. lateral $\times$ Ethnic polarization		-0.008 (0.009)	
Temp. lateral $\times$ Ethnic fractionalization			-0.017* (0.010)
Cell FE	✓	✓	✓
Country-Year FE	✓	✓	✓
Dep. Var. Mean	0.10	0.10	0.10
R ²	0.44	0.44	0.44
Cells	9,340	9,340	9,340
Observations	252,180	252,180	252,180

Notes. The table reports robustness estimates from Equation 1. Each column adds *Exposure Lateral* interacted with a different ethnic-composition measure on top of *Exposure Lateral* $\times$ *Discharge (log)*: column (1) mixed settlements; (2) ethnic polarization; (3) ethnic fractionalization (from Eberle et al. 2025). The sample is smaller than in the baseline because ethnic polarization and fractionalization are taken from the Eberle et al. (2025) replication data, whose cell coverage is slightly narrower than our panel. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B8: Robustness: upstream and downstream temperature exposure controlling for ethnic composition

	Any Conflict		
	(1)	(2)	(3)
Temp. own	-0.017** (0.008)	-0.018** (0.008)	-0.018** (0.008)
Temp. up	0.003 (0.004)	0.002 (0.004)	0.002 (0.004)
Temp. down	0.021*** (0.006)	0.022*** (0.006)	0.022*** (0.006)
Cell FE	✓	✓	✓
Country-Year FE	✓	✓	✓
Year FE	✓	✓	✓
Mixed settlements $\times$ Year FE	✓		
Ethnic polarization $\times$ Year FE		✓	
Ethnic fractionalization $\times$ Year FE			✓
Dep. Var. Mean	0.11	0.11	0.11
R ²	0.44	0.44	0.44
Cells	8,466	8,466	8,466
Observations	228,582	228,582	228,582
H ₀ : Up = Down (p)	0.03	0.02	0.02

Notes. The table reports robustness estimates from Equation 2. Each column adds year fixed effects with slopes on a different ethnic-composition measure: column (1) mixed settlements; (2) ethnic polarization; (3) ethnic fractionalization (from Eberle et al. 2025). The sample is smaller than in the baseline because ethnic polarization and fractionalization are taken from the Eberle et al. (2025) replication data, whose cell coverage is slightly narrower than our panel. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B9: Balance of upstream versus downstream neighbors' characteristics

Characteristic	Mean Up	Mean Down	N	Diff.	p -value
Population density (2000)	0.028	0.029	5,310	0.001	0.819
Urban share	0.086	0.085	5,310	-0.001	0.965
Agricultural suitability	0.244	0.237	5,310	-0.008	0.452
Travel time to urban centre	574.217	555.942	5,310	-18.274	0.278
Distance to coast (km)	715.005	669.697	5,310	-45.308	<0.001
Distance to national border	149.614	151.069	5,241	1.454	0.774
Distance to ethnic border	90.659	75.874	5,310	-14.785	<0.001
Mine present	0.026	0.022	5,286	-0.004	0.302
Gold-mining suitability	0.230	0.199	5,274	-0.030	0.016
Ethnic polarization	0.510	0.520	5,048	0.010	0.439
Ethnic fractionalization	0.448	0.442	5,048	-0.006	0.612

Notes. The table assesses whether a cell's upstream and downstream neighbors differ systematically in observable characteristics, which could offer an explanation for the up-down asymmetry other than water flow. For each on-river reference cell, and for each characteristic, we compute the mean value among its upstream neighbors (*Mean Up*) and among its downstream neighbors (*Mean Down*) within the 200 km neighborhood, averaging over neighbors in the same unweighted way as the temperature exposures. *Diff.* is the mean of the within-cell difference (down minus up) across cells; a cell enters only if it has at least one upstream and one downstream neighbor, the sample over which the upstream and downstream coefficients in column (3) of Table 2 are jointly identified. The characteristics are those used across the robustness tables. The p -value is for the test that the mean difference is zero, computed with Conley standard errors accounting for a spatial correlation of 200 km and infinite serial correlation. Distances are in kilometres; *Mine present* is an indicator, so its difference is small in absolute terms.

Table B10: Robustness: competing climatic channels

	Any Conflict					
	(1)	Lateral (2)	(3)	(4)	Up-Down (5)	(6)
Temp. own	-0.0003 (0.020)	-0.018 (0.012)	-0.0007 (0.020)	-0.026** (0.013)	-0.019** (0.008)	-0.026** (0.013)
Temp. own $\times$ Discharge (log)	-0.004 (0.004)		-0.004 (0.004)			
Temp. lateral	-0.018 (0.023)	6.45×10^{-5} (0.014)	-0.018 (0.023)	0.009 (0.014)		0.009 (0.014)
Temp. lateral $\times$ Discharge (log)	0.010** (0.004)	0.006*** (0.001)	0.010** (0.004)			
Temp. down				0.020*** (0.006)	0.021*** (0.006)	0.020*** (0.006)
Temp. up				0.004 (0.004)	0.004 (0.004)	0.004 (0.004)
Rain deficit		0.0007 (0.002)	0.0007 (0.002)		-0.0006 (0.002)	-0.0005 (0.002)
Cell FE	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.09	0.09	0.10	0.10	0.10
R ²	0.44	0.44	0.44	0.44	0.44	0.44
Cells	10,202	10,202	10,202	9,103	9,103	9,103
Observations	275,454	275,353	275,353	245,781	245,758	245,758
H ₀ : Up = Down (p)				0.03	0.03	0.03

Notes. The table reports robustness estimates for the two main specifications when controlling for competing climatic channels. Columns (1)–(3) report the lateral specification (Equation 1) on the full sample; columns (4)–(6) report the upstream–downstream specification (Equation 2) on on-river cells. Column (1) adds own-cell temperature interacted with *Discharge (log)*, allowing water richness to moderate local as well as lateral shocks; column (2) adds own-cell rainfall deficit (*Rain deficit*) as an additional control; column (3) includes both. Column (4) adds lateral temperature exposure as a control alongside upstream and downstream exposure; column (5) instead adds *Rain deficit*; column (6) includes both. The table reports the p -value for the test that upstream and downstream exposure coefficients are equal in columns (4)–(6). The columns that add the rainfall-deficit control drop 101 cell-year observations with zero precipitation, for which the log is not defined. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B11: Robustness: killing degree days as the heat shock (lateral)

	Any Conflict			
	(1)	(2)	(3)	(4)
KDD own	-0.028** (0.013)	-0.034*** (0.013)	-0.032** (0.013)	-0.034*** (0.011)
KDD lateral	0.027* (0.015)	0.013 (0.015)	0.014 (0.015)	
KDD lateral × Discharge (log)		0.008*** (0.002)		
KDD lateral × Discharge (log, annual)			0.007*** (0.002)	
Discharge (log, annual)			0.002 (0.002)	
KDD lateral water-poor				0.015* (0.009)
KDD lateral water-rich				0.0009 (0.013)
KDD lateral water-poor × Discharge (log)				0.007*** (0.003)
KDD lateral water-rich × Discharge (log)				0.0008 (0.002)
Cell FE	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.09	0.09	0.09
R ²	0.44	0.44	0.44	0.44
Cells	10,202	10,202	10,202	10,202
Observations	275,454	275,454	275,454	275,454

Notes. The table replicates Table 1 (Equation 1) using killing degree days in place of demeaned temperature. *KDD own* is the number of days with mean temperature above 32°C in the cell-year, demeaned at the cell level and scaled in units of 100 days; *KDD lateral* is the mean of the same measure in neighboring lateral cells within 200 km, excluding the cell itself (Section 3.2). Column (2) interacts lateral exposure with *Discharge (log)* computed from the sample-period mean discharge (1997–2023). Column (3) instead interacts lateral exposure with *Discharge (log, annual)* computed from cell-year mean discharge ($\log(1 + \text{discharge}_{it})$). Column (4) splits lateral exposure into water-poor and water-rich neighbors, interacted with sample-period *Discharge (log)*. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B12: Robustness: killing degree days as the heat shock (upstream–downstream)

	Any Conflict		
	(1)	(2)	(3)
KDD own	-0.027*** (0.009)	-0.016** (0.007)	-0.034*** (0.009)
KDD down	0.029*** (0.008)		0.028*** (0.008)
KDD up		0.016* (0.009)	0.015* (0.009)
Cell FE	✓	✓	✓
Country-Year FE	✓	✓	✓
Dep. Var. Mean	0.10	0.10	0.10
R ²	0.44	0.44	0.44
Cells	9,103	9,103	9,103
Observations	245,781	245,781	245,781
H ₀ : Up = Down (p)			0.27

Notes. The table replicates Table 2 (Equation 2) using killing degree days in place of demeaned temperature. *KDD own* is the number of days with mean temperature above 32°C in the cell-year, demeaned at the cell level and scaled in units of 100 days; *KDD up* and *KDD down* are the mean of the same measure in neighboring upstream and downstream cells within 200 km, excluding the cell itself (Section 3.2). Columns (1) and (2) include downstream and upstream exposure separately; column (3) includes both. The footer reports the *p*-value for the test that upstream and downstream exposure coefficients are equal. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B13: Robustness: alternative neighborhood radii (180–400 km)

	Any Conflict													
	180 km (1)	200 km (2)	250 km (3)	300 km (4)	350 km (5)	400 km (6)	200 km (IDW) (7)	180 km (8)	200 km (9)	250 km (10)	300 km (11)	350 km (12)	400 km (13)	200 km (IDW) (14)
Temp. own	-0.019 (0.012)	-0.018 (0.012)	-0.017 (0.013)	-0.017 (0.013)	-0.015 (0.012)	-0.015 (0.013)	-0.021 (0.013)	-0.019** (0.007)	-0.019** (0.008)	-0.019** (0.008)	-0.019** (0.009)	-0.018** (0.009)	-0.017* (0.009)	-0.020** (0.008)
Temp. lateral	0.0010 (0.014)	-0.0001 (0.014)	-0.0003 (0.015)	0.0002 (0.016)	-0.0002 (0.017)	0.0008 (0.018)	0.002 (0.015)							
Temp. lateral × Discharge (log)	0.006*** (0.001)	0.006*** (0.001)	0.006*** (0.001)	0.007*** (0.002)	0.007*** (0.002)	0.007*** (0.002)	0.006*** (0.001)							
Temp. up								0.004 (0.004)	0.004 (0.004)	0.003 (0.004)	0.003 (0.005)	0.003 (0.005)	0.002 (0.005)	0.005 (0.004)
Temp. down								0.020*** (0.006)	0.021*** (0.006)	0.022*** (0.007)	0.023*** (0.008)	0.023*** (0.008)	0.022*** (0.008)	0.020*** (0.006)
Cell FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.09	0.09	0.09	0.09	0.09	0.09	0.10	0.10	0.10	0.10	0.10	0.10	0.10
R ²	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44
Cells	10,202	10,202	10,202	10,202	10,202	10,202	10,202	9,103	9,103	9,103	9,103	9,103	9,103	9,103
Observations	275,454	275,454	275,454	275,454	275,454	275,454	275,454	245,781	245,781	245,781	245,781	245,781	245,781	245,781
H ₀ : Up = Down (p)								0.04	0.03	0.04	0.05	0.06	0.08	0.04

Notes. The table reports robustness to alternative neighborhood radii. Columns (1)–(7) replicate column (2) of Table 1 (Equation 1) at 180, 200, 250, 300, 350, and 400 km and weighted by distance at 200 km; columns (8)–(14) replicate column (3) of Table 2 (Equation 2) at the same radii. Neighbor exposures and Conley standard errors both use the radius stated in the column header rather than the 200 km used elsewhere; up/down/lateral classification is unchanged (Section 3.2). Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B14: Robustness: persistence in conflict and climatic shocks

	Any Conflict			
	Lateral (1)	(2)	Up-Down (3)	(4)
Temp. lateral × Discharge (log)	0.005*** (0.001)	0.004*** (0.001)		
Temp. lateral (lagged) × Discharge (log)		0.005*** (0.001)		
Temp. own	-0.013 (0.011)	-0.015 (0.012)	-0.016** (0.006)	-0.019** (0.008)
Temp. lateral	-0.002 (0.012)	0.002 (0.014)		
Temp. lateral (lagged)		-0.027*** (0.007)		
Temp. up			0.004 (0.003)	0.005 (0.003)
Temp. down			0.016*** (0.005)	0.020*** (0.006)
Temp. up (lagged)				-0.002 (0.003)
Temp. down (lagged)				0.002 (0.004)
Lagged conflict	0.192*** (0.011)		0.189*** (0.011)	
Cell FE	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.09	0.10	0.10
R ²	0.47	0.45	0.46	0.44
Cells	10,202	10,202	9,103	9,103
Observations	265,252	265,252	236,678	236,678
H ₀ : Up = Down (p)			0.05	0.08

Notes. The table reports robustness estimates addressing the temporal persistence of both conflict and climatic shocks. Columns (1)–(2) use the full sample (lateral specification, Equation 1); columns (3)–(4) restrict to on-river cells (upstream–downstream specification, Equation 2). Columns (1) and (3) augment the baseline with a one-year lag of the dependent variable (*Lagged conflict*) to address conflict persistence. Columns (2) and (4) instead augment the baseline with one-period lags of neighbor temperature exposure (*Exp. lateral lag*, *Exp. up lag*, *Exp. down lag*) to assess whether effects build over time or reflect only contemporaneous shocks. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B15: Robustness: lateral temperature exposure across discharge thresholds

	Full	p50	Any Conflict			
	(1)	(2)	p60 (3)	p70 (4)	p80 (5)	p90 (6)
Temp. own	-0.018 (0.012)	-0.005 (0.017)	-0.014 (0.015)	-0.015 (0.014)	-0.019 (0.014)	-0.020 (0.013)
Temp. lateral	-0.0001 (0.014)	-0.004 (0.018)	0.004 (0.016)	0.003 (0.016)	0.005 (0.015)	0.004 (0.014)
Temp. lateral × Discharge (log)	0.006*** (0.001)	0.006*** (0.002)	0.006** (0.003)	0.006** (0.002)	0.006*** (0.002)	0.006*** (0.002)
Cell FE	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.05	0.06	0.07	0.08	0.08
R ²	0.44	0.42	0.43	0.44	0.43	0.44
Cells	10,202	5,099	6,121	7,140	8,161	9,182
Observations	275,454	137,673	165,267	192,780	220,347	247,914

Notes. The table reports robustness of the lateral specification (Equation 1) to progressively restricting the sample to lower-discharge cells. Column (1) (*Full*) uses the entire sample; columns (2)–(6) further restrict to cells whose sample-period mean discharge is at or below the 50th, 60th, 70th, 80th, and 90th percentile, respectively, of the discharge distribution. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B16: Robustness: upstream and downstream temperature exposure across discharge thresholds

	Full	p50	Any Conflict			
	(1)	(2)	p60 (3)	p70 (4)	p80 (5)	p90 (6)
Temp. own	-0.019** (0.008)	-0.019*** (0.007)	-0.018** (0.007)	-0.017** (0.007)	-0.018** (0.007)	-0.019** (0.007)
Temp. down	0.021*** (0.006)	0.018*** (0.005)	0.017*** (0.005)	0.018*** (0.006)	0.018*** (0.006)	0.019*** (0.006)
Temp. up	0.004 (0.004)	0.006 (0.004)	0.005 (0.004)	0.002 (0.004)	0.002 (0.004)	0.002 (0.004)
Cell FE	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓
Dep. Var. Mean	0.10	0.07	0.08	0.08	0.09	0.09
R ²	0.44	0.43	0.43	0.44	0.43	0.44
Cells	9,103	4,548	5,459	6,372	7,282	8,193
Observations	245,781	122,796	147,393	172,044	196,614	221,211
H ₀ : Up = Down (p)	0.03	0.07	0.08	0.04	0.02	0.02

Notes. The table reports robustness of the upstream–downstream specification (Equation 2) to progressively restricting the sample to lower-discharge cells. Column (1) (*Full*) uses the entire on-river sample; columns (2)–(6) further restrict to cells whose sample-period mean discharge is at or below the 50th, 60th, 70th, 80th, and 90th percentile, respectively, of the on-river discharge distribution. The table reports the p -value for the test that upstream and downstream exposure coefficients are equal. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B17: Robustness: alternative water-richness measures and on-river sample definitions

	Any Conflict			
	Lateral (1)	(2)	Up-Down (3)	(4)
Temp. own	-0.018 (0.012)	-0.018 (0.012)	-0.020** (0.008)	-0.021** (0.009)
Temp. lateral	0.007 (0.014)	0.007 (0.014)		
Temp. lateral $\times$ Discharge (log)	0.004*** (0.0010)			
Temp. lateral $\times$ Water rich		0.031*** (0.009)		
Temp. up			0.004 (0.004)	0.003 (0.005)
Temp. down			0.022*** (0.006)	0.024*** (0.007)
Cell FE	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓
Dep. Var. Mean	0.10	0.09	0.10	0.11
R ²	0.44	0.44	0.44	0.44
Cells	9,552	10,202	8,837	7,272
Observations	257,904	275,454	238,599	196,344
H ₀ : Up = Down (p)			0.03	0.02

Notes. The table reports robustness estimates for the two main specifications under alternative operationalizations of water richness and of the on-river sample. Columns (1)–(2) report the lateral specification (Equation 1) on the full sample, interacting *Temp. lateral* with two alternative measures of water richness: column (1) uses *Discharge (log)*, here the log of mean cell discharge over 1997–2023, which is defined only for cells with positive discharge; column (2) uses *Water rich*, an indicator equal to one for cells whose sample-period mean discharge is above the sample median. Columns (3)–(4) report the upstream–downstream specification (Equation 2) under two stricter definitions of the on-river sample: a cell is classified as on-river if it has positive mean discharge and either perennial rivers or seasonal river length above the 25th (column 3) or 50th (column 4) percentile of the seasonal-river length distribution. The table reports the *p*-value for the test that upstream and downstream exposure coefficients are equal in columns (3)–(4). Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B18: Robustness: UCDP GED data

	Any Conflict (GED)		State Conflict (GED)		Non-State Conflict (GED)	
	(1)	(2)	(3)	(4)	(5)	(6)
Temp. own	-0.017*	-0.010*	-0.012	-0.008	-0.010	-0.007*
	(0.009)	(0.005)	(0.008)	(0.005)	(0.007)	(0.004)
Temp. lateral	0.013		0.009		0.005	
	(0.011)		(0.010)		(0.008)	
Temp. lateral × Discharge (log)	0.001		0.0009		0.001*	
	(0.001)		(0.001)		(0.0006)	
Temp. up		-0.0003		0.001		-0.002
		(0.003)		(0.002)		(0.002)
Temp. down		0.010***		0.007**		0.006**
		(0.004)		(0.003)		(0.003)
Cell FE	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓
Dep. Var. Mean	0.04	0.04	0.03	0.03	0.02	0.03
R ²	0.33	0.33	0.32	0.32	0.29	0.29
Cells	10,202	9,103	10,202	9,103	10,202	9,103
Observations	275,454	245,781	275,454	245,781	275,454	245,781
H ₀ : Up = Down (p)		0.03		0.16		0.02

Notes. The table reports robustness estimates using the same specifications as Tables 1 and 2, but with UCDP GED conflict incidence as the dependent variable. Odd-numbered columns—(1), (3), and (5)—replicate the lateral specification from Equation 1 on the full sample; even-numbered columns—(2), (4), and (6)—replicate the upstream–downstream specification from Equation 2 on on-river cells. Columns (1)–(2) use any GED conflict incidence; columns (3)–(4) use state-based GED conflict incidence; columns (5)–(6) use non-state GED conflict incidence. The table reports the p -value for the test that upstream and downstream exposure coefficients are equal in columns (2), (4), and (6). Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B19: Robustness: lateral temperature exposure, alternative inference

	Any Conflict							Cell + Country	Cell + climate zone-year
	200 km	250 km	300 km	350 km	400 km	450 km	500 km		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Temp. own	-0.018	-0.018	-0.018	-0.018	-0.018	-0.018	-0.018	-0.018	-0.018**
	(0.012)	(0.012)	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.008)
Temp. lateral	-0.0001	-0.0001	-0.0001	-0.0001	-0.0001	-0.0001	-0.0001	-0.0001	-0.0001
	(0.014)	(0.014)	(0.015)	(0.015)	(0.016)	(0.016)	(0.016)	(0.022)	(0.011)
Temp. lateral × Discharge (log)	0.006***	0.006***	0.006***	0.006***	0.006***	0.006***	0.006***	0.006***	0.006***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)
Cell FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.09	0.09	0.09	0.09	0.09	0.09	0.09	0.09
R ²	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44
Cells	10,202	10,202	10,202	10,202	10,202	10,202	10,202	10,202	10,202
Observations	275,454	275,454	275,454	275,454	275,454	275,454	275,454	275,454	275,454

Notes. The table replicates column (2) of Table 1 under alternative inference. Columns (1)–(7) report Conley spatial-HAC standard errors with infinite serial correlation at the cell level and spatial correlation cutoffs from 200 km to 500 km. Column (8) two-way clusters standard errors at the cell and country level; column (9) two-way clusters at the cell and climate zone-year level. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B20: Robustness: upstream and downstream temperature exposure, alternative inference

	Any Conflict								Cell + climate zone-year (9)
	200 km (1)	250 km (2)	300 km (3)	350 km (4)	400 km (5)	450 km (6)	500 km (7)	Cell + Country (8)	
Temp. own	-0.019** (0.008)	-0.019** (0.008)	-0.019** (0.009)	-0.019** (0.009)	-0.019** (0.009)	-0.019** (0.009)	-0.019** (0.010)	-0.019* (0.011)	-0.019*** (0.006)
Temp. up	0.004 (0.004)	0.004 (0.004)	0.004 (0.004)	0.004 (0.005)	0.004 (0.005)	0.004 (0.005)	0.004 (0.005)	0.004 (0.004)	0.004 (0.004)
Temp. down	0.021*** (0.006)	0.021*** (0.006)	0.021*** (0.007)	0.021*** (0.007)	0.021*** (0.007)	0.021*** (0.007)	0.021*** (0.007)	0.021** (0.010)	0.021*** (0.005)
Cell FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Dep. Var. Mean	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10
R ²	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44
Cells	9,103	9,103	9,103	9,103	9,103	9,103	9,103	9,103	9,103
Observations	245,781	245,781	245,781	245,781	245,781	245,781	245,781	245,781	245,781
H ₀ : Up = Down (p)	0.03	0.04	0.05	0.05	0.06	0.06	0.06	0.14	0.0007

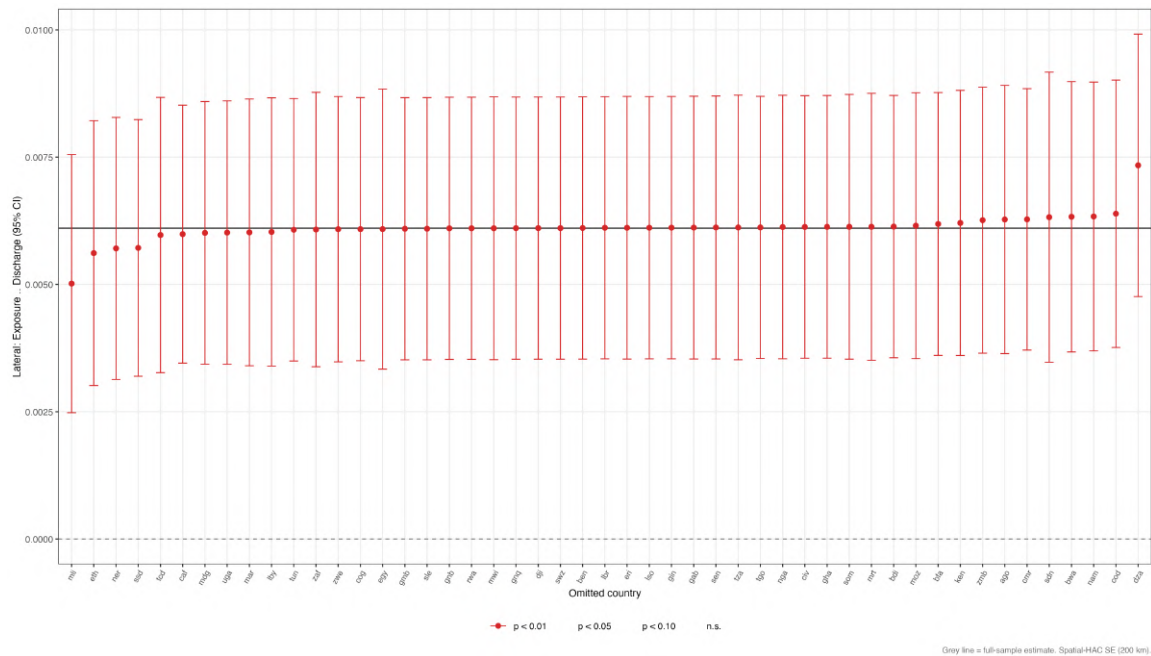
Notes. The table replicates column (3) of Table 2 under alternative inference, on the on-river sample. Columns (1)–(7) report Conley spatial-HAC standard errors with infinite serial correlation at the cell level and spatial correlation cutoffs from 200 km to 500 km. Column (8) two-way clusters standard errors at the cell and country level; column (9) two-way clusters at the cell and climate zone-year level. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B21: Robustness: baseline specifications on the language-coded subsample

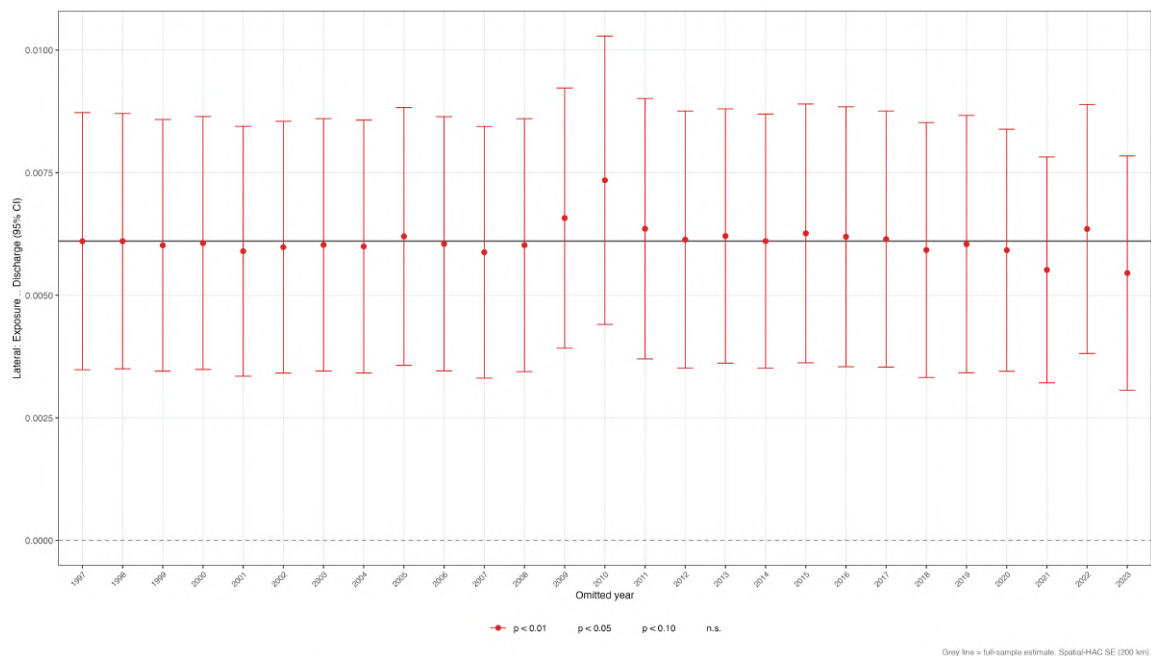
Sample	Any Conflict			
	Lateral (H1)		Up-Down (H2)	
	Full (1)	Language-coded (2)	Full (3)	Language-coded (4)
Temp. own	-0.018 (0.012)	-0.019 (0.013)	-0.019** (0.008)	-0.021** (0.008)
Temp. lateral	-0.0001 (0.014)	0.003 (0.015)		
Temp. lateral × Discharge (log)	0.006*** (0.001)	0.005*** (0.001)		
Temp. up			0.004 (0.004)	0.001 (0.004)
Temp. down			0.021*** (0.006)	0.025*** (0.007)
Cell FE	✓	✓	✓	✓
Country-Year FE	✓	✓	✓	✓
Dep. Var. Mean	0.09	0.10	0.10	0.11
R ²	0.44	0.44	0.44	0.44
Cells	10,202	8,687	9,103	8,093
Observations	275,454	234,549	245,781	218,511
H ₀ : Up = Down (p)			0.03	<0.01

Notes. The table re-estimates the two baseline specifications on the subsample of cells with a non-missing Ethnologue language, the sample used for the same- versus other-ethnic-group decomposition in columns 2 and 4 of Table 5. Columns (1)–(2) report the lateral specification of column (2) of Table 1 (Equation 1) on the full sample and on the language-coded cells. Columns (3)–(4) report the upstream-downstream specification of column (3) of Table 2 (Equation 2), restricted to on-river cells, on the full on-river sample and on the language-coded on-river cells. The cells without a language assignment are non-random, concentrated in sparsely populated arid margins outside the mapped ethno-linguistic areas. The footer reports the p -value for the test that upstream and downstream exposure coefficients are equal. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Figure B1 – Robustness: leave-one-out jackknife for lateral exposure interacted with discharge



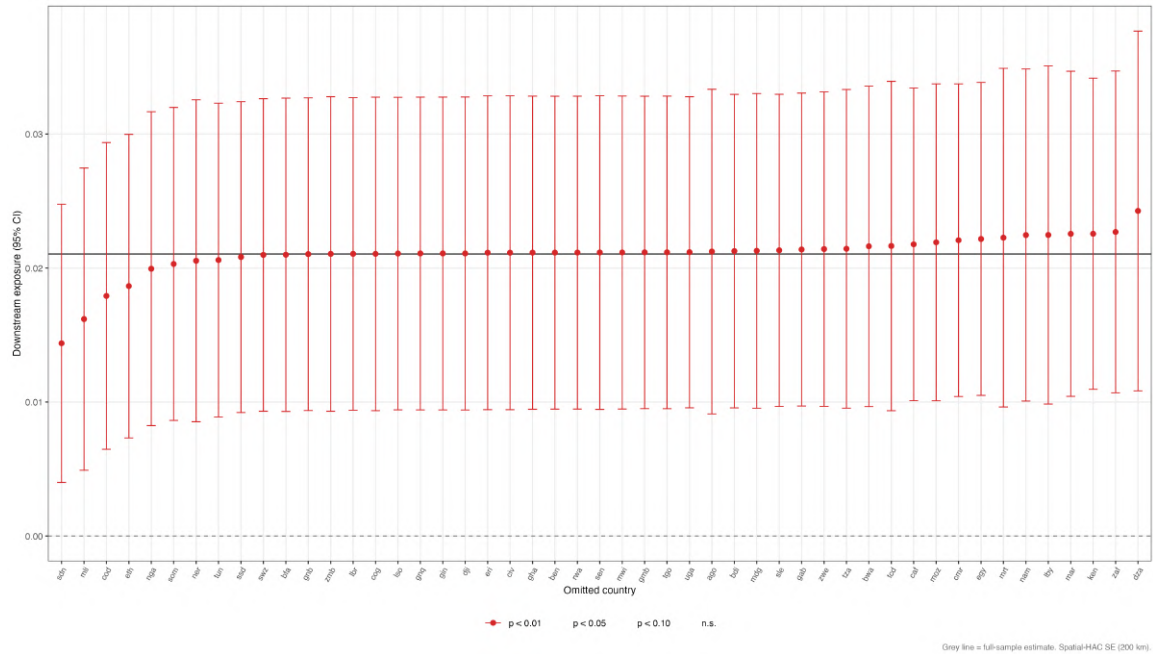
(a) Leave-one-country-out



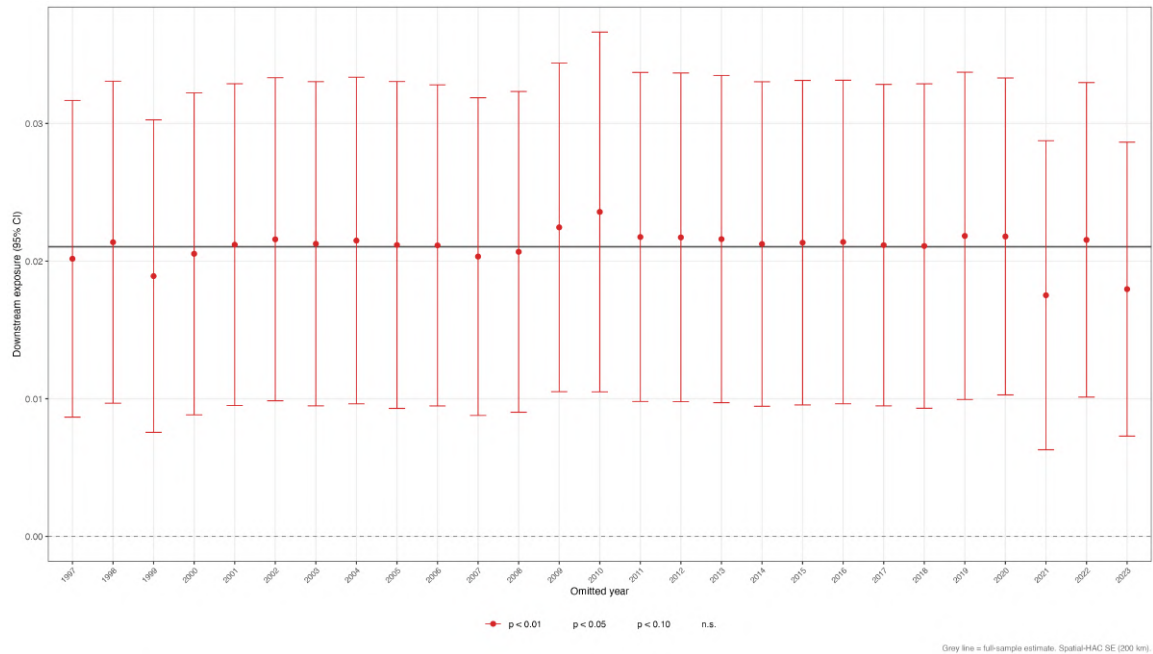
(b) Leave-one-year-out

Notes. Each panel re-estimates column (2) of Table 1 (Equation 1) while excluding all observations from one country (top) or one year (bottom). The dependent variable is any ACLED conflict incidence in a cell-year. The reported coefficient is $Exposure\ Lateral \times Discharge\ (log)$. Points show coefficient estimates and 95% confidence intervals; colors indicate statistical significance. The horizontal grey line is the full-sample estimate. Conley spatial-HAC standard errors with a 200 km cutoff and infinite serial correlation at the cell level.

Figure B2 – Robustness: leave-one-out jackknife for downstream temperature exposure



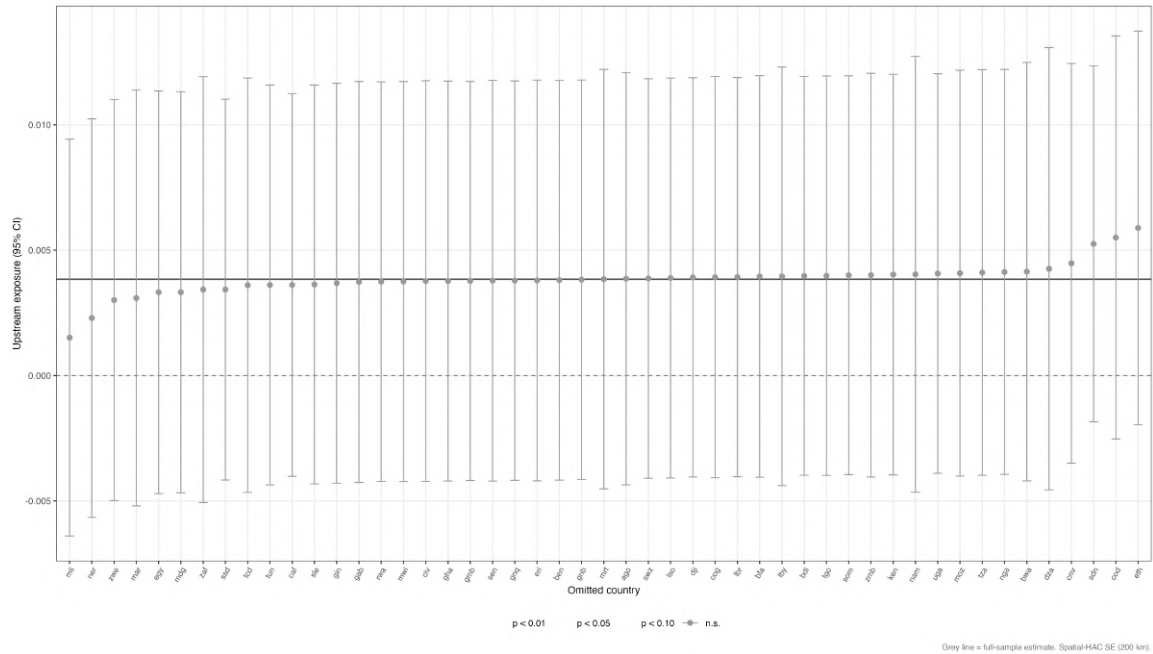
(a) Leave-one-country-out



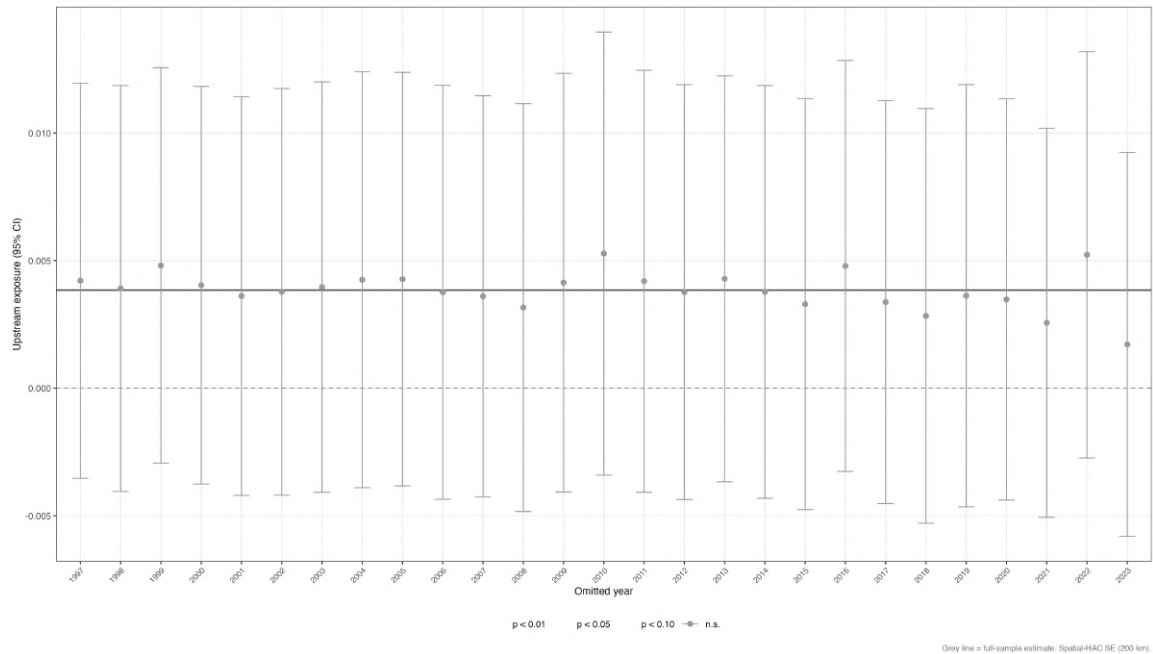
(b) Leave-one-year-out

Notes. Each panel re-estimates column (3) of Table 2 (Equation 2) on the on-river sample while excluding all observations from one country (top) or one year (bottom). The dependent variable is any ACLED conflict incidence in a cell-year. The reported coefficient is *Exposure Down*. Points show coefficient estimates and 95% confidence intervals; colors indicate statistical significance. The horizontal grey line is the full-sample estimate. Conley spatial-HAC standard errors with a 200 km cutoff and infinite serial correlation at the cell level.

Figure B3 — Robustness: leave-one-out jackknife for upstream temperature exposure



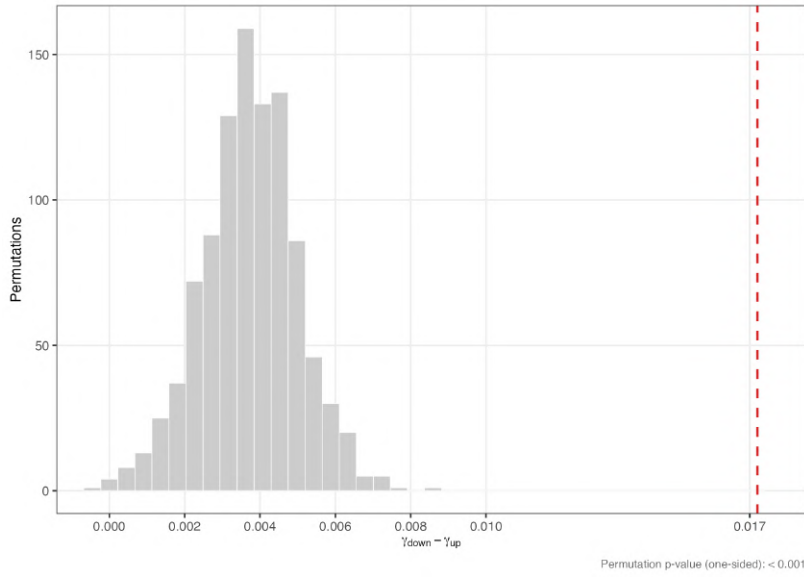
(a) Leave-one-country-out



(b) Leave-one-year-out

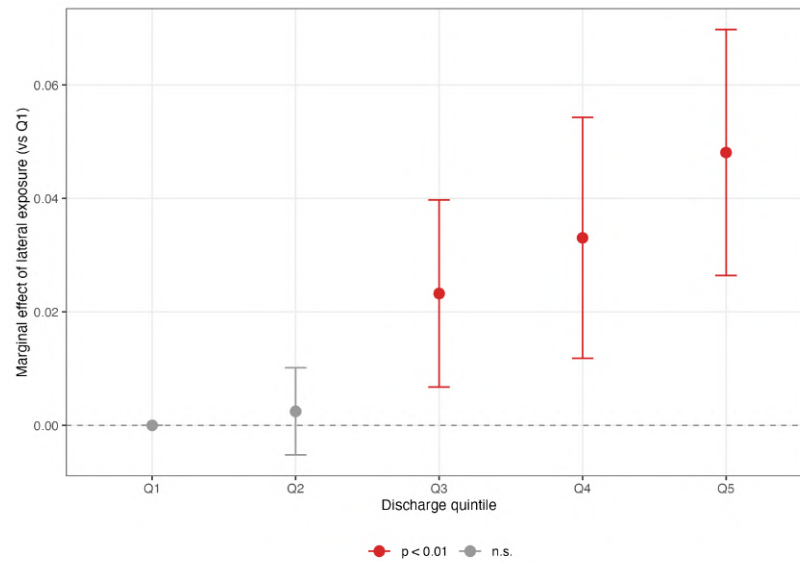
Notes. Each panel re-estimates column (3) of Table 2 (Equation 2) on the on-river sample while excluding all observations from one country (top) or one year (bottom). The dependent variable is any ACLED conflict incidence in a cell-year. The reported coefficient is *Exposure Up*. Points show coefficient estimates and 95% confidence intervals; colors indicate statistical significance. The horizontal grey line is the full-sample estimate. Conley spatial-HAC standard errors with a 200 km cutoff and infinite serial correlation at the cell level.

Figure B4 – Permutation test of downstream–upstream asymmetry



Notes: The figure reports a restricted permutation test of the up–down asymmetry in column (3) of Table 2. Under the sharp null that the direction of flow is irrelevant, the labelling of a cell’s river neighbors as upstream or downstream is uninformative. We test this by permuting the upstream and downstream labels among each reference cell’s directional neighbors, holding fixed the neighbors and their temperatures and preserving each cell’s own count of upstream and downstream neighbors, and re-estimating the difference $\gamma_{\text{down}} - \gamma_{\text{up}}$ for each of 1,000 relabelings. The histogram is the resulting null distribution; the vertical line marks the observed difference. The observed asymmetry lies beyond the entire null distribution. The null is centred slightly above zero because upstream and downstream are not symmetric positions in the river network, which makes the test conservative.

Figure B5 — Effect of lateral temperature by discharge quintiles



Notes: The figure plots the marginal effect of lateral temperature exposure (*Exposure Lateral*) on any ACLED conflict incidence, allowing the effect to vary by quintile of the reference cell's *Discharge (log)* (mean discharge over 1997–2023). The unit of observation is a $0.5^\circ \times 0.5^\circ$ grid cell and year; the full sample is used (not restricted to on-river cells). Points show the interaction of *Exposure Lateral* with each discharge-quintile indicator, relative to the lowest-discharge quintile (Q1, normalized to zero); vertical lines are 95% confidence intervals. Colors indicate statistical significance. All columns include cell fixed effects and country-year fixed effects. Conley spatial-HAC standard errors with a 200 km cutoff and infinite serial correlation at the cell level.